

ETHICS IN AI: EMPHASIS ON FAIRNESS

CSC-391 Technical Communication Report

by

GROUP 7

Aryan Choudhary	23115024
Nitin Agiwal	23114074
Pradyuman Singh Shekhawat	23115107
Siddharth Gupta	23114093
Vishesh Gupta	23114107

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY ROORKEE
ROORKEE – 247 667 (INDIA)
APRIL, 2026

Contribution Statement

Name	Contribution
Nitin Agiwal	Studied and analyzed the problem of bias and fairness in AI systems. Reviewed foundational concepts including representation learning, Variational Autoencoders (VAEs), and adversarial learning, and contributed to framing the core problem statement.
Pradyuman Singh Shekhawat	Conducted the literature review with a focus on the FD-VAE architecture. Explained the fairness mechanisms underlying three-way latent disentanglement and the role of the decorrelation module in the representation learning stage.
Siddharth Gupta	Contributed to the literature review and helped design the proposed fairness-aware pipeline, with special focus on integrating Fractional Fourier Transform preprocessing and adversarial debiasing into the overall architecture.
Aryan Choudhary	Contributed to the literature review and collaborated on the design of the proposed pipeline, with particular focus on the CVaR fairness loss formulation and its two variants: demographic-agnostic (DAG) and demographic-aware (DAW).
Vishesh Gupta	Interpreted the fairness evaluation metrics including Equal Opportunity, Equalized Odds, and Equalized Accuracy. Analyzed the experimental results on the CelebA and UTK Face datasets and connected the technical findings to real-world AI systems and societal implications.

Table 1: Individual contributions of team members

Acknowledgements

This report was prepared as part of the course *CSC-391: Technical Communication* at the Indian Institute of Technology Roorkee. We express our sincere gratitude to the course instructors, Prof. Raghvendra Singh Rohit and Prof. Chetan Gupta, for designing an assignment that encouraged not only technical rigor but also critical synthesis across diverse research directions. Their guidance played a crucial role in shaping both the structure and depth of this work.

We also extend our appreciation to the Department of Computer Science and Engineering and the Institute administration, including the Director of the Indian Institute of Technology Roorkee, for fostering an academic environment that supports interdisciplinary exploration and responsible engagement with emerging technologies such as artificial intelligence.

The theoretical foundation of this report is built upon prior research contributions. In particular, the FD-VAE framework proposed by Park, Kim, Hwang, and Byun (2020) provided the core representational perspective, while the work of Ju, Hu, Jia, Chen, and Lyu (2024) informed the integration of Conditional Value-at-Risk (CVaR) optimization and fairness-centric evaluation strategies in deepfake detection. We acknowledge these authors, as well as the broader research and open-source community, whose publicly available datasets—CelebA, Celeb-DF, DFD, and DFDC—and tools such as IBM’s AI Fairness 360 enabled meaningful empirical analysis.

The five members of Group 7 contributed complementary strengths to this project, including mathematical analysis, literature synthesis, system design, and contextual interpretation of results. The collaborative process, characterized by continuous feedback and critical discussion, significantly enhanced the quality and coherence of the final report.

Disclaimer: This report is an academic exercise prepared for the CSC-391 course and does not represent original research. The content is based on a synthesis of existing literature and publicly available resources, and any errors or misinterpretations are solely the responsibility of the authors.

Abstract

The past decade has witnessed an unprecedented expansion of artificial intelligence from narrowly scoped research prototypes into production systems that inform, and in many cases determine, outcomes in domains as consequential as medical diagnosis, credit allocation, recruitment, and criminal sentencing. This growth has brought with it a class of problems that purely technical performance metrics are ill-equipped to capture: the risk that AI systems reproduce, amplify, or systematically worsen pre-existing social inequities. Among the many dimensions of AI ethics that have emerged in response, transparency, accountability, privacy, and safety, *fairness* occupies a particularly central position, both because its violation is directly measurable and because its consequences fall disproportionately on already-marginalised populations.

This report investigates fairness in AI with a sustained focus on the problem of demographic bias in machine-learning pipelines. Bias in learned systems is not typically the product of deliberate design choices; it enters at multiple stages of the development pipeline, including the collection of training data that reflects historical inequalities, the underrepresentation of minority groups in benchmark datasets, and the subjective choices made during feature engineering and label annotation. Once embedded in a model's parameters, such bias is difficult to detect through accuracy-based evaluation alone, because aggregate performance statistics mask disparate error rates across demographic subgroups.

To address these shortcomings, the report surveys and synthesises a family of fairness-aware machine-learning methods centred on *representation learning*, the paradigm of learning compact, structured feature encodings from raw data. In particular, attention is directed toward *disentangled* representation learning, which seeks to decompose a learned latent space into independent factors corresponding to distinct attributes of the data. By explicitly separating protected demographic information from task-relevant predictive information at the level of the latent representation, downstream classifiers can be shielded from sensitive attributes without sacrificing predictive utility.

The report examines the theoretical foundations of this approach, including Variational Autoencoders (VAEs) and their fairness-aware extension, the FD-VAE (Fairness-aware Disentangling VAE), which introduces a three-way partition of the latent space into target-attribute, protected-attribute, and mutual-attribute subspaces. It further surveys

adversarial debiasing strategies, wherein a gradient reversal layer forces the encoder to produce representations from which a demographic classifier cannot reliably recover group membership. Complementing these representational approaches, *Conditional Value-at-Risk* (CVaR) is examined as a distributionally robust training objective that focuses optimisation on worst-performing demographic subgroups, thereby improving cross-group consistency even in the absence of explicit demographic labels.

The report also presents a task-specific pipeline combining Fractional Fourier Transform (FrFT) pre-processing, adversarial autoencoding, latent consistency regularisation, and a CVaR-based fairness loss, designed with deepfake detection as the target application. This pipeline is evaluated using a comprehensive suite of fairness metrics, including Equal Opportunity, Equalized Odds, Equalized Accuracy, and FPR-based disparity measures, which together provide a multi-faceted view of demographic parity.

The overarching aim of this work is to demonstrate that accuracy and fairness are not fundamentally in tension, and that fairness-aware representation learning offers a principled path toward AI systems that are simultaneously reliable, equitable, and socially responsible.

Keywords: fairness in AI, bias mitigation, representation learning, adversarial debiasing

Contents

Acknowledgements	v
Abstract	vii
List of Figures	xiv
List of Tables	xv
List of Abbreviations	xviii
Structure of Report	xix
1 Introduction	1
1.1 AI in Society: Scale and Stakes	1
1.2 Core Ethical Principles	2
1.3 Bias as a Systemic Challenge	3
1.4 Real-World Case Studies	4
1.4.1 COMPAS: Algorithmic Bias in Criminal Justice	4
1.4.2 Amazon's Recruiting Tool: Gender Bias in Hiring	5
1.5 Why Accuracy Metrics Are Insufficient	5
1.6 Motivation for Fairness-Aware Learning	6
1.6.1 Limitations of Prior Approaches	6
2 Preliminaries, Background, and Evaluation Metrics	7
2.1 Mathematical Foundations	7
2.1.1 Sets, Functions, and Notation	7
2.1.2 Infimum and Supremum	8
2.1.3 Kullback-Leibler Divergence	8
2.1.4 Total Correlation	8
2.1.5 Mutual Information	9
2.1.6 Evidence Lower Bound (ELBO)	9
2.1.7 Conditional Value-at-Risk (CVaR)	9
2.2 Representation Learning	10
2.3 Disentangled Representation Learning	11
2.4 Autoencoders	12

2.5	Variational Autoencoders (VAE)	12
2.6	Adversarial Learning	13
2.6.1	Generative Adversarial Networks (GANs)	13
2.6.2	Adversarial Debiasing	13
2.7	Fractional Fourier Transform (FrFT)	14
2.8	Fairness Concepts in Machine Learning	15
2.9	Evaluation Metrics	16
2.9.1	Equal Opportunity	16
2.9.2	Equalized Odds	16
2.9.3	Standard Accuracy	16
2.9.4	Equalized Accuracy	16
2.9.5	FPR-Based Fairness Metrics for Deepfake Detection	17
2.9.6	Summary of Metrics	17
2.10	Key Deep Learning Architectures Used in Fair AI	17
2.11	Chapter Summary	19
3	README Framework: Fair Representation Learning Through Disentanglement	20
3.1	Motivation and Problem Setting	20
3.1.1	Limitation of Earlier Disentanglement Approaches	21
3.2	Overview of the FD-VAE Architecture	21
3.2.1	The Three Latent Subspaces	22
3.3	The Variational Autoencoder Objective	23
3.4	Disentanglement Loss via Total Correlation	24
3.4.1	Adversarial Approximation of the Discriminator	24
3.5	Decorrelation Loss	24
3.5.1	Classification Loss L_{CLS}	25
3.5.2	Adversarial Loss L_{ADV}	25
3.5.3	The Role of the Mutual Attribute Latent in Decorrelation	25
3.5.4	Total Objective for Stage 1	26
3.6	Downstream Classification Network	26
3.7	Datasets	27
3.8	Experimental Results	27
3.8.1	Results on CelebA	27
3.8.2	Results on UTK Face	28
3.9	Summary	29
4	CVaR-Based Fair Deepfake Detection: DAG-FDD and DAW-FDD	30
4.1	Introduction	30
4.2	Background: Why Average Loss Is Not Enough	31

4.3	CVaR as the Core Robustness Principle	31
4.3.1	Interpretation of the variables	32
4.3.2	Intuition	32
4.4	DAG-FDD: Demographic-Agnostic Fair Deepfake Detection	32
4.4.1	Motivation	32
4.4.2	Latent-Group Formulation	32
4.4.3	Why CVaR Helps in the Demographic-Agnostic Setting	33
4.4.4	Empirical Objective Used in DAG-FDD	33
4.4.5	Optimisation of DAG-FDD	33
4.4.6	Interpretation	34
4.5	DAW-FDD: Demographic-Aware Fair Deepfake Detection	34
4.5.1	Motivation	34
4.5.2	Group Risk	34
4.5.3	Group-Level CVaR	34
4.5.4	Per-Group Top k / CVaR Estimation	35
4.5.5	Final DAW-FDD Objective	36
4.6	Relationship Between DAG-FDD and DAW-FDD	36
4.7	Experimental Results Reported in the Paper	37
4.7.1	Key Observations	37
4.7.2	Results on FF++ (In-Domain Analysis)	38
4.7.3	Cross-Dataset Generalisation	38
4.7.4	Backbone Robustness	40
4.7.5	Computational Considerations	40
4.8	Key Insights for Our Project	40
4.9	Conclusion	41
5	Targeting a Critical Gap in Deepfake Detectors	42
5.1	Problem Statement	42
5.2	Overview of the FAIR-FRFT Architecture	43
5.3	Fractional Fourier Transform Pre-processing	43
5.4	Adversarial Autoencoder	44
5.5	Adversarial Demographic Debiasing via Gradient Reversal	45
5.6	CVaR Fairness Loss	46
5.7	Model Pipeline	46
5.8	Chapter Summary	47
6	Conclusion, Future Work, and Industry Approaches	48
6.1	Conclusion	48
6.2	Future Work	49
6.3	How AI Industry Leaders Address Fairness	50

6.3.1	OpenAI , Reinforcement Learning with Human Feedback (RLHF) .	50
6.3.2	Google , Responsible AI Policies	50
6.3.3	IBM , AI Fairness 360 (AIF360)	51
6.3.4	Comparative Summary	51
Bibliography		53



List of Figures

1.1	The five core ethical principles of AI	2
1.2	The four stages of the ML pipeline at which demographic bias typically enters, and the dominant mechanism at each stage.	3
1.3	Overall accuracy can hide uneven performance across demographic groups.	5
2.1	Comparison of sample weighting under ERM and CVaR at $\alpha = 0.3$. ERM weights all samples uniformly, whereas CVaR assigns zero weight to easy samples ($\text{loss} < \lambda^*$) and amplified weight to hard samples ($\text{loss} > \lambda^*$). Minimising CVaR therefore focuses training on the worst-performing subgroups.	10
2.2	High-level picture of representation learning. The encoder f_θ maps a raw input $\mathbf{x}$ to a structured latent vector $\mathbf{z}$, which is then used for a downstream task such as classification or fairness-aware prediction.	11
2.3	Illustration of disentangled representation learning showing mapping from observable data to feature space and factors of variation.	11
2.4	Autoencoder architecture: the encoder g_ϕ maps input $\mathbf{x}$ to a low-dimensional latent representation $\mathbf{z}$ (bottleneck), and the decoder f_θ reconstructs the input as $\mathbf{x}' \approx \mathbf{x}$	12
2.5	Generative Adversarial Network (GAN): the generator maps latent variables to synthetic samples, while the discriminator distinguishes between real and generated data, providing a training signal via adversarial loss.	14
2.6	Fractional Fourier Transform (FRFT) representations at different fractional orders α , showing the rotation of signal energy in the time-frequency plane.	15
3.1	Motivation for fairness-aware disentanglement. (a) Prior methods (e.g., FFVAE) mix target and protected information, causing loss of useful features. (b) README separates them into TAL, PAL, and MAL, isolating shared information without harming performance.	22
3.2	Full architecture of the Fairness-aware Disentangling VAE (FD-VAE). Source: adapted from Park <i>et al.</i> [1].	22
3.3	Downstream classification network of FD-VAE. Source: adapted from Park <i>et al.</i> [1].	27

4.1	Comparison of fairness-aware methods on FF++ using the Xception backbone (adapted from [2]).	38
4.2	Evaluation on Celeb-DF, DFD, and DFDC using Xception (adapted from [2]). Results show fairness and detection metrics across datasets.	38
4.3	Intersectional fairness comparison of the Xception detector across four datasets: FF++, Celeb-DF, DFD, and DFDC (adapted from Figure 1 in [2]).	39
4.4	FPR comparison of the Xception detector on intersectional groups for FF++ and DFDC (adapted from Figure 2 in [2]). Orange and dark cyan bars denote the groups with the highest and lowest FPR, respectively, and the red bracket shows the gap, where a smaller value is better.	39
4.5	Performance across multiple backbones (ResNet-50, EfficientNet-B3, DSP-FWA, RECCE) on FF++ (adapted from [2]).	40
5.1	FAIR-FRFT Architecture. Top: FrFT-transformed input x' is encoded by E_θ into latent z , reconstructed by G_θ to $\hat{x}$, and re-encoded for consistency. The model includes demographic adversary D_ϕ , real/fake head R_ψ , and CVaR fairness loss. Bottom: After training, the encoder is frozen and a classifier predicts real/fake labels from the learned fair representation. . . .	44

List of Tables

1	Individual contributions of team members	iii
2.1	Summary of fairness and utility evaluation metrics used in this report. All fairness metrics are minimised; accuracy metrics are maximised.	18
2.2	Key model architectures referenced in the fairness-in-AI literature covered in this report.	18
3.1	Classification results on the CelebA dataset. M = male, Y = young, A = attractive, W = wavy hair, B = big nose. PA = protected attribute, TA = target attribute. ↓ indicates lower is better (fairer); ↑ indicates higher is better. Bold entries indicate the best result in each column. Source: Park <i>et al.</i> [1].	28
3.2	Classification results on the UTK Face dataset. G = gender (protected), E = ethnicity (target), A = age (target). ↓ = lower is better; ↑ = higher is better. Bold = best result. Source: Park <i>et al.</i> [1].	28
4.1	Comparison of DAG-FDD and DAW-FDD.	36
6.1	Comparison of industry approaches to AI fairness.	52

List of Abbreviations

ACC	Accuracy
AE	Autoencoder
AUC	Area Under the Curve
Celeb-DF	Celeb DeepFake Dataset
COMPAS	Correctional Offender Management Profiling for Alternative Sanctions
CVaR	Conditional Value-at-Risk
DAG	Demographic-Agnostic
DAG-FDD	Demographic-Agnostic Fair Deepfake Detection
DAW	Demographic-Aware
DAW-FDD	Demographic-Aware Fair Deepfake Detection
DFD	DeepFakeDetection Dataset
DFDC	DeepFake Detection Challenge Dataset
DRL	Disentangled Representation Learning
DRO	Distributionally Robust Optimization
ELBO	Evidence Lower Bound
EO	Equal Opportunity
EOdd	Equalized Odds
FD-VAE	Fairness-aware Disentangling Variational Autoencoder
FDD	Fair Deepfake Detection
FEO	Equalized Odds-based Fairness Metric

FF++	FaceForensics++
FFPR	False Positive Rate Disparity (Fairness FPR)
FFVAE	Fairness-aware Disentangling Variational Autoencoder (prior method)
FPR	False Positive Rate
FrFT	Fractional Fourier Transform
GAN	Generative Adversarial Network
GFPR	Group False Positive Rate
GRL	Gradient Reversal Layer
KL	Kullback–Leibler Divergence
MAL	Mutual Attribute Latent
PAL	Protected Attribute Latent
TAL	Target Attribute Latent
TC	Total Correlation
TPR	True Positive Rate
VAE	Variational Autoencoder

Structure of Report

The report is organised as follows:

Chapter 1 introduces the motivation for studying ethics in AI, explains why fairness is an important concern, and outlines the main ideas addressed in the report.

Chapter 2 presents the background material needed for the rest of the discussion. It covers the mathematical and machine learning concepts used in the report, along with the fairness metrics that are later used for evaluation.

Chapter 3 discusses the ReadME paper in detail, focusing on its architecture, main ideas, and results. This chapter helps establish the basis for the methods studied in the rest of the report.

Chapter 4 presents the DAG-FDD and DAW-FDD approaches and explains their architecture and performance. These methods are examined as important fairness-aware baselines for the problem under study.

Chapter 5 describes our proposed architecture to tackle a very specific yet important gap. It explains the design choices, the overall pipeline, and the role of each component in improving fairness while preserving predictive performance.

Finally, Chapter 6 concludes the report by summarising the main findings and discussing possible directions for future work. It also reviews how major AI companies are approaching fairness and safety in practice. This chapter gives a broader view of the methods and policies currently being used in real-world systems.

Chapter 1

Introduction

Artificial intelligence has moved, in a remarkably short span of time, from the edges of academic research into the operational core of institutions that shape daily life. Decisions about who receives a loan, who advances past an automated resume screen, how a medical scan is flagged for review, and whether a defendant is assessed as a recidivism risk are increasingly mediated by learned models. This transition has delivered genuine efficiency gains, but it has also surfaced a set of ethical challenges that the field was, for much of its early history, poorly equipped to address. Foremost among these is the question of fairness: whether the errors that AI systems make are distributed uniformly across the population, or concentrated disproportionately among groups that are already disadvantaged.

This report examines the theoretical and algorithmic foundations of fairness-aware machine learning, surveys the architectural approaches that have emerged in response to demographic bias, and presents a pipeline that combines several of these approaches for fair deepfake detection. The chapter proceeds from context and motivation through real-world case studies to a discussion of methodological limitations and an overview of the report structure.

1.1 AI in Society: Scale and Stakes

The contemporary wave of AI is distinguished from earlier periods principally by scale, of data, compute, and deployment. Deep learning models that once struggled to classify handwritten digits can now generate realistic images, translate across dozens of languages, and diagnose pathologies from medical images with accuracy comparable to specialist physicians. These capabilities have diffused rapidly into industry and government. Recent surveys suggest that over three-quarters of organisations worldwide have incorporated AI into at least some operational processes [3].

The domains of deployment are increasingly consequential. Algorithmic credit scoring evaluates loan applications at a scale and speed that human review cannot match. Hiring platforms use natural language processing to filter thousands of resumes before a recruiter

sees any candidate. Law enforcement agencies in several jurisdictions have deployed facial recognition and recidivism prediction tools with direct implications for individual liberty. Healthcare systems use predictive models to triage patients and flag diagnostic anomalies.

What distinguishes these applications from earlier automation is a shift from *explicit rule specification* to *implicit statistical learning*. A rule-based system encodes its logic transparently and can be inspected and revised. A deep neural network trained on historical data encodes its decision logic in millions of numerical parameters, learning from data that may carry the imprint of historical inequalities. Any biases present in the training distribution are likely to be reproduced, and can be amplified, in the model’s predictions, without any deliberate intention on the part of the system’s designers.

1.2 Core Ethical Principles

The AI ethics literature has converged on a relatively stable set of principles. Figure 1.1 summarises the five most widely cited. This report focuses on fairness, but a brief account of the others establishes necessary context.

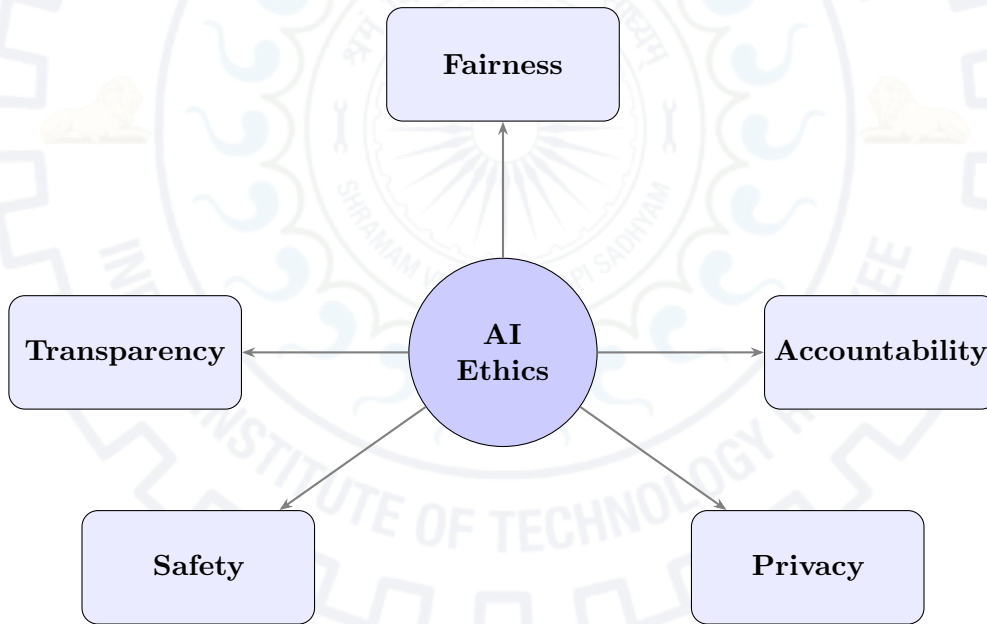


Figure 1.1: The five core ethical principles of AI

Accountability requires identifiable agents, individuals, teams, or organisations, to be responsible for an AI system’s behaviour and answerable when that behaviour causes harm. In practice, accountability is complicated by the distributed nature of AI development: the organisation that collects training data often differs from the one that trains the model, which may differ again from the one that deploys it.

Transparency and explainability concern making AI processes and individual decisions legible to those they affect. Explainability has particular legal significance in contexts such as credit lending, where applicants may have a right to reasons for an adverse

decision.

Privacy is threatened by the same richness of learned representations that makes AI useful: detailed internal models of data distributions can leak sensitive personal information. Privacy-preserving techniques such as differential privacy and federated learning partially address this risk.

Safety demands that AI systems behave reliably, including under distribution shift and adversarial manipulation. In safety-critical domains such as autonomous driving or medical diagnosis, the consequences of failure can be catastrophic and irreversible.

Fairness, the focus of this report, is, broadly, the principle that a system should not treat individuals or groups inequitably on the basis of morally irrelevant or legally protected characteristics. This deceptively simple statement conceals technical complexity: multiple formally distinct fairness criteria exist (demographic parity, equal opportunity, equalized odds, individual fairness), and they are, in general, mutually incompatible [4]. The choice among them is not a purely technical decision but a normative one that depends on the values and context of the application.

1.3 Bias as a Systemic Challenge

A common misconception is that AI bias is primarily a data quality problem: if the training data were clean and representative, the model would be fair. While data quality matters, this framing understates the depth of the challenge. Bias enters the pipeline at multiple stages, and addressing it requires interventions at each.

Figure 1.2 illustrates the four stages at which bias typically enters.

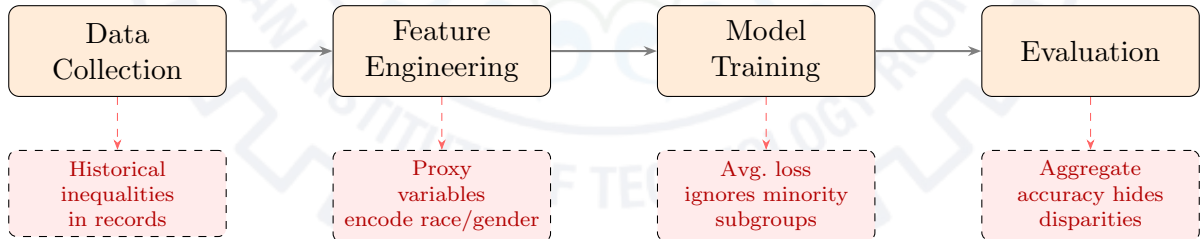


Figure 1.2: The four stages of the ML pipeline at which demographic bias typically enters, and the dominant mechanism at each stage.

Data collection. Historical records, hiring decisions, criminal justice outcomes, medical referrals, reflect the actions of agents operating within social structures shaped by unequal treatment. A model trained on such data learns to conflate genuine predictive relationships with the legacy of past discrimination.

Feature engineering. Even when legally protected attributes such as race and gender are explicitly excluded from the feature set, proxy variables, postal code, university attended, surname, may carry substantial demographic information, allowing the model to act on the very attributes that were ostensibly suppressed.

Model training. Standard loss functions minimise average error, treating all examples equally. Because minority subgroups constitute a small fraction of the training population, a model can achieve a low average loss while performing substantially worse on those subgroups than on the majority. This *accuracy disparity* is not a bug in the training algorithm; it is a predictable consequence of optimising a metric that does not account for distributional heterogeneity.

Evaluation. Aggregate accuracy conceals subgroup disparities. A model with 95% accuracy on a majority group representing 80% of the test set and 70% accuracy on a minority group representing 20% yields an overall accuracy of $(0.8 \times 0.95) + (0.2 \times 0.70) = 0.90$, an impressive headline figure that obscures a 25-percentage-point gap. The same limitation applies to AUC: a model can rank majority-group members reliably while placing minority-group members near the decision boundary, yet still report a high aggregate AUC.

1.4 Real-World Case Studies

The concerns outlined above are not hypothetical. Two well-documented cases illustrate the concrete consequences of bias in high-stakes AI systems.

1.4.1 COMPAS: Algorithmic Bias in Criminal Justice

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) algorithm is used in several American states to estimate a defendant’s likelihood of reoffending. Judges and parole boards may consult its risk scores when deciding bail, sentencing, and supervision. A 2016 investigation by ProPublica found substantial racial disparities: Black defendants were nearly twice as likely as White defendants to be incorrectly classified as high risk, with a false positive rate of approximately 45% compared to 23% for White defendants [5].

Three features of this case are instructive. First, race is not a direct input to COMPAS; the disparities arise because input features such as prior arrest records and socioeconomic survey responses are themselves correlated with race, reflecting historical patterns of policing. Second, the developers could credibly claim the system was calibrated, within each risk-score band, observed reoffending rates were roughly equal across racial groups, illustrating the fundamental incompatibility between calibration and equal false-positive-rate criteria when base rates differ across groups [4]. Third, the consequences of an incorrect high-risk classification are severe and difficult to remedy: a person may be held in custody or receive a harsher sentence.

1.4.2 Amazon’s Recruiting Tool: Gender Bias in Hiring

A major technology company developed an automated resume screening system trained on ten years of historical hiring decisions. Because the workforce those decisions produced was heavily male in technical roles, the model learned to penalise signals associated with female applicants: resumes mentioning graduates of all-women’s colleges were downgraded, as were those containing any language with gendered connotations. The system was abandoned after the bias was discovered [6].

The case illustrates how machine learning can perpetuate historical inequalities even when protected attributes are excluded from the feature set, and how the critical question of *what is being predicted* goes unanswered. A system trained on past hiring decisions measures *resemblance to historically successful employees* rather than *ability to succeed in the role*, a distinction with profound fairness implications that is invisible in the training signal.

1.5 Why Accuracy Metrics Are Insufficient

The two cases discussed earlier reflect a broader methodological issue: standard classification metrics are not well suited to detect or characterise demographic bias.

Overall accuracy is a natural metric when prediction errors carry equal cost and the test distribution matches real-world deployment. However, these assumptions rarely hold in high-stakes applications. False positives and false negatives often have asymmetric consequences, and datasets are frequently imbalanced across demographic groups.

The deeper limitation is that accuracy is an *aggregate metric*. Aggregation masks variations in performance across subgroups. As a result, significant disparities can remain hidden behind strong overall accuracy figures. A model may appear effective globally while performing poorly for specific populations.

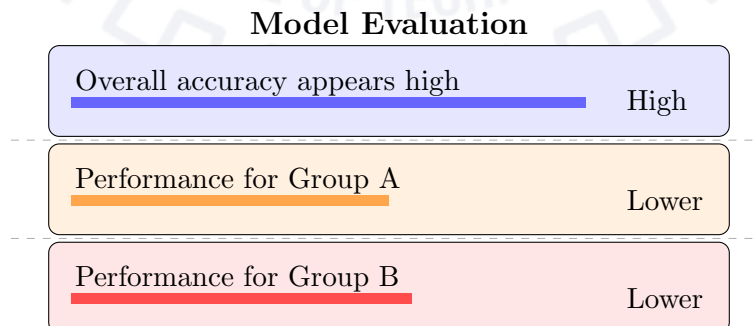


Figure 1.3: Overall accuracy can hide uneven performance across demographic groups.

To address this, fairness metrics explicitly measure cross-group performance differences. Metrics such as *Equal Opportunity*, *Equalized Odds*, *Equalized Accuracy*, and *FPR-based disparity* provide a more nuanced understanding of model behaviour. These metrics are

formally defined in Chapter 2 and are used throughout this report for evaluation.

1.6 Motivation for Fairness-Aware Learning

Recognising that bias is systemic and that aggregate metrics are insufficient motivates the need for fairness-aware design. Traditional approaches often rely on post-hoc corrections, such as calibration or output reweighting. While useful, these methods are inherently reactive and cannot address biases embedded within learned representations.

A more principled approach focuses on the representations themselves. If the features learned by a model do not encode protected attributes, then predictions derived from these features are less likely to exhibit systematic bias. This idea forms the basis of *fair representation learning*, which aims to balance predictive utility with demographic invariance.

However, achieving this balance is challenging. In real-world data, target and protected attributes are often correlated. For example, factors such as employment and housing stability may influence prediction tasks while also being correlated with demographic characteristics. Removing all demographic information risks discarding useful predictive signals.

This trade-off between fairness and performance represents a central challenge in modern machine learning. The goal is to reduce bias without compromising accuracy, a problem that motivates the methods explored in this report.

1.6.1 Limitations of Prior Approaches

Existing fairness methods are typically categorised based on where the intervention occurs. *Pre-processing* methods modify the data before training, for instance by rebalancing datasets. *In-processing* methods introduce fairness constraints directly into the learning objective. *Post-processing* methods adjust predictions after the model has been trained.

Each category has inherent limitations. Pre-processing may remove informative features, reducing model performance. Post-processing cannot eliminate bias embedded in internal representations. In-processing methods, if not carefully designed, may fail to account for complex dependencies between attributes.

A further limitation arises in disentangled representation learning. Earlier approaches, such as FFVAE, partitioned the latent space into “sensitive” and “non-sensitive” components. However, due to correlations between attributes, demographic information can still be indirectly encoded in the non-sensitive subspace.

The FD-VAE framework addresses this limitation by introducing an additional *mutual* latent space that captures shared information between target and protected attributes.

Chapter 2

Preliminaries, Background, and Evaluation Metrics

Declaration of Contributions:

- **Siddharth Gupta** — was responsible for the sections on representation learning, including autoencoders, variational autoencoders, disentangled representations, and related architectural foundations.
 - **Aryan Choudhary** — contributed the development and formalisation of the Conditional Value-at-Risk (CVaR) framework and its role in fairness-aware optimisation.
 - **Vishesh Gupta** — designed and documented the evaluation metrics, including fairness and performance measures such as Equal Opportunity, Equalized Odds, and FPR-based metrics.
-

This chapter lays the conceptual and mathematical groundwork required to understand the fairness-aware machine-learning techniques surveyed in subsequent chapters. We begin with the essential mathematical building blocks—drawing on analysis and probability—then introduce the AI/ML architectures that appear throughout the literature, and finally define the fairness metrics used to quantify bias.

2.1 Mathematical Foundations

2.1.1 Sets, Functions, and Notation

Throughout this report we adopt the following notation. Scalars are denoted by lower-case italic letters (x, y, z); vectors by bold lower-case ($\mathbf{x}, \mathbf{z}$); and matrices by bold

upper-case $(\mathbf{W}, \mathbf{X})$. The set of real numbers is $\mathbb{R}$, and $\mathbb{R}^d$ denotes a d -dimensional real vector space. We write

$\mathcal{X}$ for an input space and $\mathcal{Z}$ for a latent (hidden) space.

A *random variable* X defined over a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ takes values in $\mathcal{X}$. Its *probability distribution* is denoted P_X or simply P . The *expectation* of a function $f(X)$ is

$$\mathbb{E}_{X \sim P}[f(X)] = \int_{\mathcal{X}} f(x) \, dP(x).$$

2.1.2 Infimum and Supremum

Two operations appear repeatedly in the loss functions of the papers we review.

- The **infimum** (inf) of a set $S \subseteq \mathbb{R}$ is its *greatest lower bound*: the largest value ℓ such that $s \geq \ell$ for every $s \in S$. For a function $f : \Lambda \rightarrow \mathbb{R}$ we write $\inf_{\lambda \in \Lambda} f(\lambda)$ to mean the smallest value f can approach (not necessarily attained).
- The **supremum** (sup) is the *least upper bound*, analogously defined.

These concepts are essential when the minimum or maximum of a function is not attained—a situation that arises naturally in risk-based fairness objectives (see Section 2.1.7).

2.1.3 Kullback-Leibler Divergence

Given two probability distributions P and Q over the same space, the *Kullback–Leibler (KL) divergence* measures how much P diverges from Q :

$$D_{\text{KL}}(P \| Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right] = \int P(x) \log \frac{P(x)}{Q(x)} \, dx. \quad (2.1)$$

$D_{\text{KL}}(P \| Q) \geq 0$ always, with equality iff $P = Q$ almost everywhere. KL divergence is *asymmetric*: in general $D_{\text{KL}}(P \| Q) \neq D_{\text{KL}}(Q \| P)$.

Role in VAEs. In Variational Autoencoders (Section 2.5), the KL divergence acts as a regulariser that pushes the approximate posterior $q_{\Phi}(\mathbf{z} | \mathbf{x})$ towards the standard Gaussian prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$.

2.1.4 Total Correlation

Total Correlation (TC), introduced by Watanabe [7], generalises mutual information to measure the overall statistical dependency among n random variables $z_1, z_2, \dots, z_n$:

$$\text{TC}(z_1, \dots, z_n) = D_{\text{KL}} \left(q(z_1, z_2, \dots, z_n) \parallel \prod_{j=1}^n q(z_j) \right). \quad (2.2)$$

When TC equals zero, all variables are mutually independent. Minimising TC during training therefore encourages a model to learn representations whose components carry non-overlapping information—the core idea behind *disentangled* representation learning (Section 2.3).

2.1.5 Mutual Information

The *mutual information* between two random variables X and Y quantifies how much knowing X reduces uncertainty about Y :

$$I(X; Y) = D_{\text{KL}}(P_{XY} \| P_X \otimes P_Y) = \mathbb{E}_{(x,y) \sim P_{XY}} \left[\log \frac{P_{XY}(x, y)}{P_X(x) P_Y(y)} \right]. \quad (2.3)$$

In fair representation learning, we want the encoded features $\mathbf{z}$ to have *high* mutual information with the input $\mathbf{x}$ (to preserve utility) but *low* mutual information with a sensitive attribute s (to ensure fairness).

2.1.6 Evidence Lower Bound (ELBO)

Exact marginal likelihood $\log p_{\Theta}(\mathbf{x})$ is intractable for deep latent-variable models. Jensen’s inequality gives a tractable surrogate known as the **Evidence Lower Bound (ELBO)**:

$$\log p_{\Theta}(\mathbf{x}) \geq \underbrace{\mathbb{E}_{q_{\Phi}(\mathbf{z}|\mathbf{x})}[\log p_{\Theta}(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_{\Phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{regularisation term}} =: \mathcal{L}_{\text{VAE}}. \quad (2.4)$$

Maximising $\mathcal{L}_{\text{VAE}}$ simultaneously encourages faithful reconstruction and a well-behaved latent distribution.

2.1.7 Conditional Value-at-Risk (CVaR)

Standard empirical risk minimisation (ERM) minimises the *average* loss:

$$\mathcal{R}_{\text{avg}}(\theta) := \mathbb{E}_{(X,Y) \sim P}[\ell(\theta; X, Y)]. \quad (2.5)$$

Because the average is dominated by easy (majority) samples, ERM tends to underperform on rare demographic subgroups—precisely the fairness problem. *Distributionally Robust Optimisation* (DRO) addresses this by minimising the *worst-case* risk $\mathcal{R}_{\text{max}}(\theta) = \max_m \mathbb{E}_{(X,Y) \sim P_m}[\ell(\theta; X, Y)]$ over unknown latent groups.

A tractable upper bound on $\mathcal{R}_{\text{max}}$ is given by *Conditional Value-at-Risk* at level

$\alpha \in (0, 1)$ [8]:

$$\text{CVaR}_\alpha(\theta) := \inf_{\lambda \in \mathbb{R}} \left\{ \lambda + \frac{1}{\alpha} \mathbb{E}_{(X,Y) \sim P} [\ell(\theta; X, Y) - \lambda]_+ \right\}, \quad (2.6)$$

where $[a]_+ = \max\{0, a\}$ is the hinge (ReLU) function.

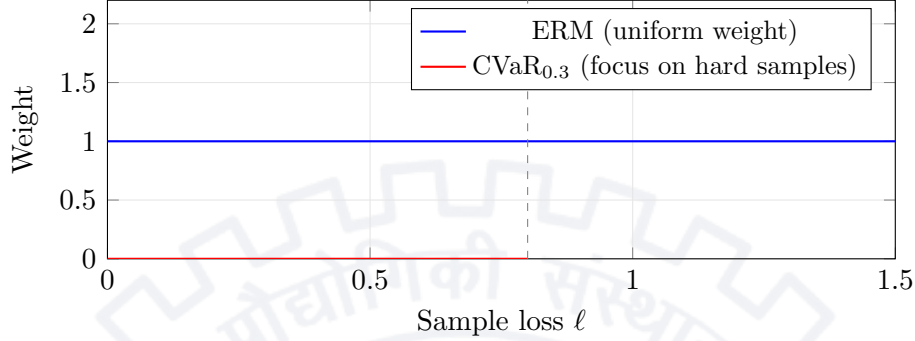


Figure 2.1: Comparison of sample weighting under ERM and CVaR at $\alpha = 0.3$. ERM weights all samples uniformly, whereas CVaR assigns zero weight to easy samples (loss $< \lambda^*$) and amplified weight to hard samples (loss $> \lambda^*$). Minimising CVaR therefore focuses training on the worst-performing subgroups.

Proposition (Proposition 1 in [9]). *If each latent group m occurs with probability $\pi_m \geq \alpha$, then $\text{CVaR}_\alpha(\theta) \geq \mathcal{R}_{\max}(\theta)$. Hence minimising CVaR minimises an upper bound on the worst-case group risk.*

Group-level CVaR. When demographic labels are available, one can apply CVaR at the group level:

$$\text{CVaR}_\alpha^{\mathcal{G}}(\theta) := \inf_{\lambda \in \mathbb{R}} \left\{ \lambda + \frac{1}{\alpha} \mathbb{E}_{G \sim \text{Uniform}(\mathcal{G})} [\mathcal{R}_G(\theta) - \lambda]_+ \right\}, \quad (2.7)$$

where $\mathcal{R}_G(\theta)$ is the per-group risk. This formulation simultaneously minimises average group risk and minimises the disparity across groups [10].

2.2 Representation Learning

Representation learning is the subfield of machine learning concerned with automatically discovering compact, informative features (representations) from raw data [11]. Rather than hand-crafting features, a model—typically a deep neural network—learns a function $f_\theta : \mathcal{X} \rightarrow \mathcal{Z}$ that maps high-dimensional inputs $\mathbf{x}$ to a lower-dimensional *latent* (embedding) vector $\mathbf{z}$. The key desiderata are:

1. **Informativeness:** $\mathbf{z}$ retains enough information about $\mathbf{x}$ for downstream tasks.
2. **Compactness:** $\mathbf{z}$ discards irrelevant variation (noise, style, sensitive attributes).
3. **Transferability:** representations learned on one task generalise to related tasks.

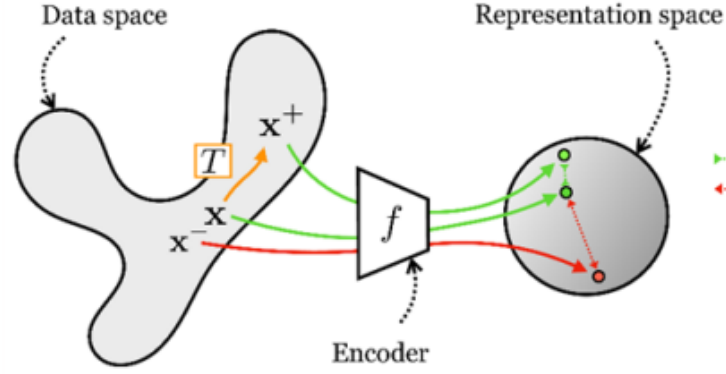


Figure 2.2: High-level picture of representation learning. The encoder f_θ maps a raw input $\mathbf{x}$ to a structured latent vector $\mathbf{z}$, which is then used for a downstream task such as classification or fairness-aware prediction.

2.3 Disentangled Representation Learning

Standard representation learning makes no assumption about the *structure* of the latent space. *Disentangled* representation learning imposes the additional constraint that different dimensions (or subspaces) of $\mathbf{z}$ capture statistically independent generative factors of the data [11, 12, 13].

Formally, if the data is generated by K independent factors $v_1, v_2, \dots, v_K$, a disentangled encoder should produce a latent vector $\mathbf{z} = [z_1, z_2, \dots, z_K]$ such that

$$q_\Phi(z_1, z_2, \dots, z_K) \approx \prod_{k=1}^K q_\Phi(z_k),$$

i.e., the joint latent distribution factorises into independent marginals. This is precisely the condition that Total Correlation (Eq. 2.2) is zero.

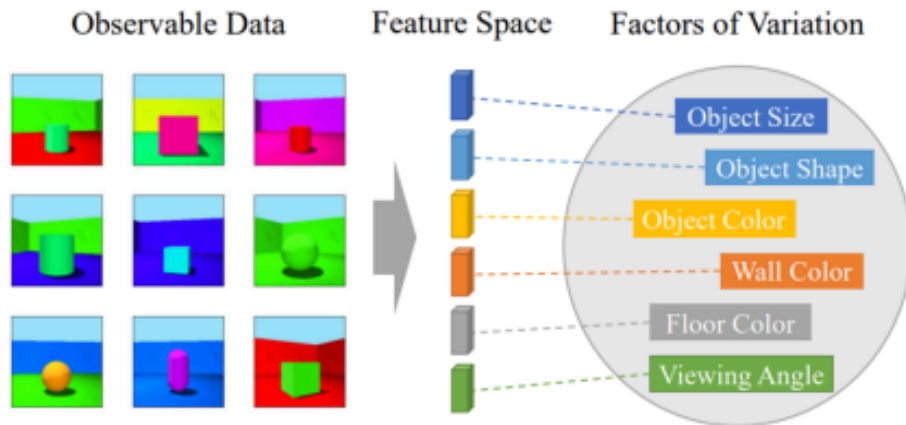


Figure 2.3: Illustration of disentangled representation learning showing mapping from observable data to feature space and factors of variation.

Why does disentanglement matter for fairness? If a latent space is properly disentangled, one can isolate and *remove* the subspace that encodes sensitive demographic information before passing representations to a downstream classifier, achieving fairness without sacrificing task-relevant information.

2.4 Autoencoders

An *autoencoder* is a neural network trained to reconstruct its input through a bottleneck layer. It consists of two components:

- **Encoder** $E_\theta : \mathcal{X} \rightarrow \mathcal{Z}$ compresses the input $\mathbf{x}$ into a latent code $\mathbf{z} = E_\theta(\mathbf{x})$.
- **Decoder** $G_\phi : \mathcal{Z} \rightarrow \mathcal{X}$ reconstructs the input as $\hat{\mathbf{x}} = G_\phi(\mathbf{z})$.

Training minimises reconstruction loss, typically mean-squared error:

$$\mathcal{L}_{\text{AE}} = \mathbb{E}_{\mathbf{x}} \left[\|\mathbf{x} - G_\phi(E_\theta(\mathbf{x}))\|^2 \right]. \quad (2.8)$$

The bottleneck forces the encoder to discard redundant information and retain only the structure necessary for reconstruction.

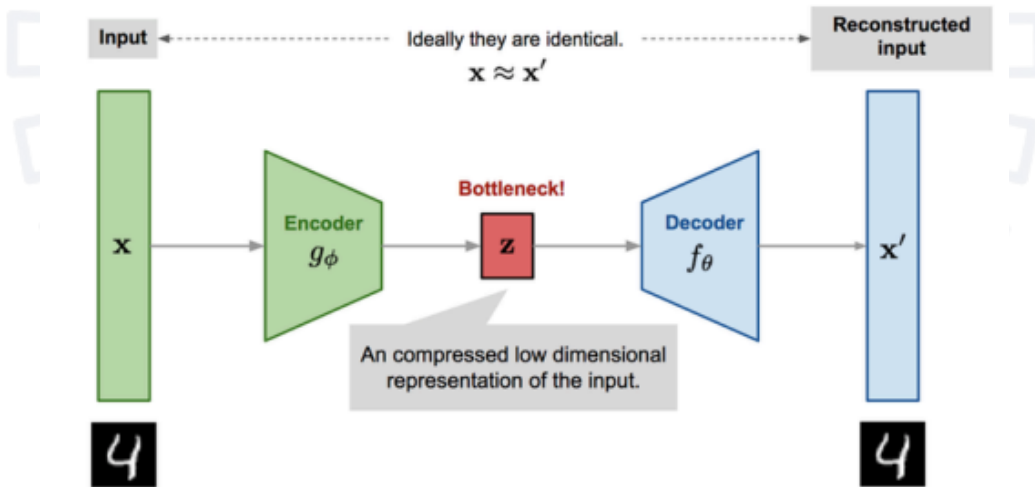


Figure 2.4: Autoencoder architecture: the encoder g_ϕ maps input $\mathbf{x}$ to a low-dimensional latent representation $\mathbf{z}$ (bottleneck), and the decoder f_θ reconstructs the input as $\mathbf{x}' \approx \mathbf{x}$.

2.5 Variational Autoencoders (VAE)

A *Variational Autoencoder* (VAE) [14] is a probabilistic generative model that frames encoding and decoding as Bayesian inference. Instead of mapping an input to a single point in latent space, the encoder outputs a *distribution*:

$$q_\Phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_{q_\Phi}(\mathbf{x}), \sigma_{q_\Phi}^2(\mathbf{x})),$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are output by the encoder network with parameters Φ . The decoder models $p_{\Theta}(\mathbf{x}|\mathbf{z})$.

Reparameterisation trick. To enable gradient-based optimisation through a stochastic node, we sample

$$\mathbf{z} = \boldsymbol{\mu}_{q_{\Phi}}(\mathbf{x}) + \boldsymbol{\sigma}_{q_{\Phi}}(\mathbf{x}) \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (2.9)$$

which moves the randomness outside the computational graph.

Training objective. The VAE is trained by maximising the ELBO (Eq. 2.4), which can be written explicitly as:

$$\mathcal{L}_{\text{VAE}} = \sum_{i=1}^N \left[\mathbb{E}_{q_{\Phi}(\mathbf{z}|\mathbf{x}_i)} [\log p_{\Theta}(\mathbf{x}_i|\mathbf{z})] - D_{\text{KL}}(q_{\Phi}(\mathbf{z}|\mathbf{x}_i) \parallel p(\mathbf{z})) \right]. \quad (2.10)$$

Extensions for disentanglement. β -VAE [12] up-weights the KL term by a hyperparameter $\beta > 1$ to encourage greater disentanglement at the cost of reconstruction quality. FactorVAE [13] and β -TCVAE [15] instead minimise Total Correlation directly (Eq. 2.2), yielding better disentanglement without degrading reconstruction.

2.6 Adversarial Learning

Adversarial learning is a training paradigm in which two model components compete in a minimax game, each driving the other towards a more useful solution [16].

2.6.1 Generative Adversarial Networks (GANs)

In the canonical GAN setup, a generator G learns to produce synthetic samples that fool a discriminator D , while D learns to distinguish real from fake:

$$\min_G \max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (2.11)$$

At equilibrium, G produces samples from p_{data} and D cannot distinguish real from generated samples.

2.6.2 Adversarial Debiasing

In fair representation learning, adversarial networks serve a different role. A *predictor* P is trained to predict a target label y from a representation $\mathbf{z}$, while an *adversary* A attempts to predict the sensitive attribute s from the same $\mathbf{z}$. The encoder is then trained to *fool the adversary*:

$$\min_{E,P} \max_A \mathcal{L}_{\text{pred}}(P(E(\mathbf{x})), y) - \lambda \mathcal{L}_{\text{adv}}(A(E(\mathbf{x})), s). \quad (2.12)$$

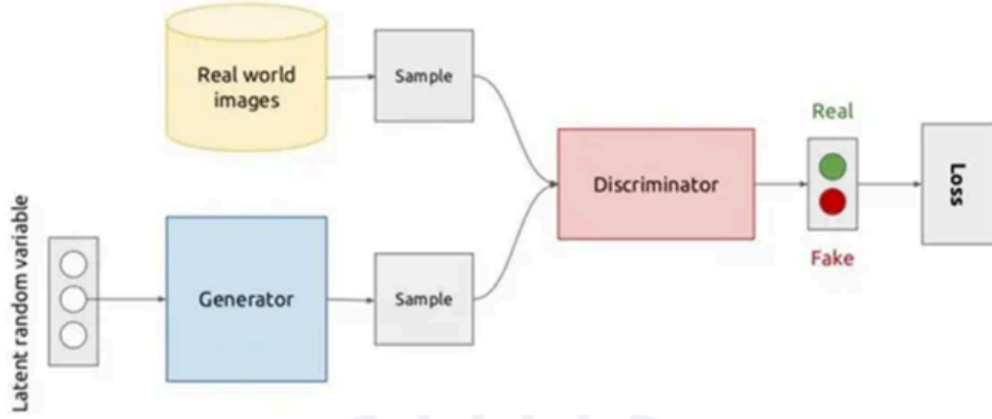


Figure 2.5: Generative Adversarial Network (GAN): the generator maps latent variables to synthetic samples, while the discriminator distinguishes between real and generated data, providing a training signal via adversarial loss.

The negative sign on the adversarial loss means the encoder *maximises* the adversary’s error—equivalently, it removes sensitive information from $\mathbf{z}$.

Gradient Reversal Layer (GRL). A practical implementation of Eq. 2.12 uses a Gradient Reversal Layer (GRL) [17]. During the forward pass, GRL acts as the identity; during back-propagation, it negates the gradient, automatically implementing the minimax objective without alternating optimisation:

$$\text{GRL}(\mathbf{z}) = \mathbf{z}, \quad \frac{\partial \text{GRL}(\mathbf{z})}{\partial \mathbf{z}} = -\lambda_{\text{grl}} \mathbf{I}. \quad (2.13)$$

2.7 Fractional Fourier Transform (FrFT)

The classical Fourier transform maps a signal from the time domain to the frequency domain. The *Fractional Fourier Transform* (FrFT) [18] is a generalisation parameterised by a rotation order $\alpha \in [0, 1]$, interpolating continuously between the time ($\alpha = 0$) and frequency ($\alpha = 1$) domains:

$$\mathcal{F}_\alpha[f](u) = \int_{-\infty}^{\infty} f(t) K_\alpha(t, u) dt, \quad (2.14)$$

where the kernel is

$$K_\alpha(t, u) = A_\alpha \exp(i\pi(t^2 \cot \alpha - 2tu \csc \alpha + u^2 \cot \alpha)), \quad A_\alpha = \sqrt{1 - i \cot \alpha}.$$

Deepfake generation pipelines introduce subtle frequency artefacts that are more discriminative in intermediate spatial–frequency domains than in either the pure spatial or pure frequency domain [19]. By selecting α between 0 and 1, the FrFT exposes these artefacts while reducing reliance on demographic appearance features (skin tone, facial structure) that dominate the spatial domain.

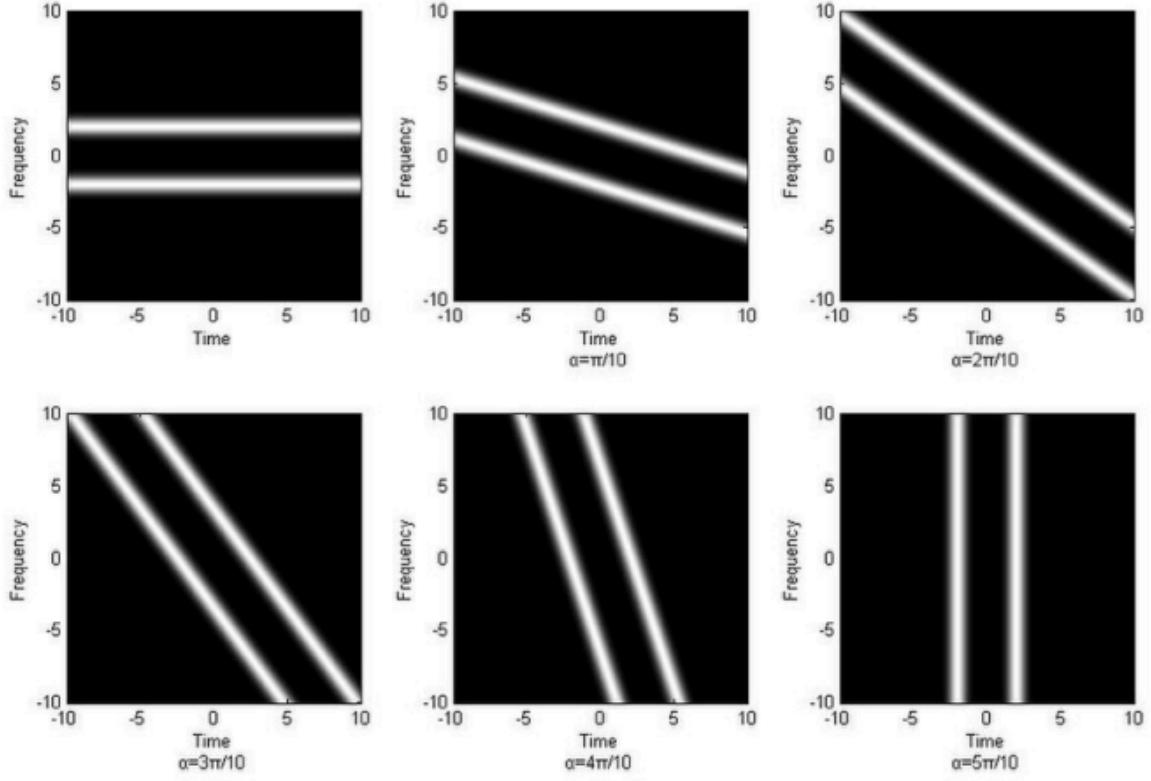


Figure 2.6: Fractional Fourier Transform (FRFT) representations at different fractional orders α , showing the rotation of signal energy in the time-frequency plane.

2.8 Fairness Concepts in Machine Learning

Before defining evaluation metrics, we establish the terminology used throughout this report.

- **Target attribute** (y_t): the label the model is trained to predict (e.g., “attractive”, “real/fake”).
- **Protected (sensitive) attribute** (y_p): a demographic characteristic (e.g., gender, race, age) with respect to which the model should *not* discriminate.
- **Group fairness**: equal treatment of pre-specified demographic groups.
- **Individual fairness**: similar individuals receive similar predictions.

Let $\hat{Y}$ denote the model’s prediction, Y the ground truth label ($Y = 1$ positive, $Y = 0$ negative), and $G \in \{g_0, g_1\}$ the protected group membership. Define:

$$\text{TPR}_g = \Pr(\hat{Y} = 1 \mid Y = 1, G = g), \quad (\text{True Positive Rate for group } g)$$

$$\text{TNR}_g = \Pr(\hat{Y} = 0 \mid Y = 0, G = g), \quad (\text{True Negative Rate for group } g)$$

$$\text{FPR}_g = \Pr(\hat{Y} = 1 \mid Y = 0, G = g). \quad (\text{False Positive Rate for group } g)$$

2.9 Evaluation Metrics

2.9.1 Equal Opportunity

Equal Opportunity [20] requires that the True Positive Rate be equal across protected groups. A model satisfies equal opportunity if subjects who *deserve* a positive outcome are equally likely to receive it, regardless of group membership. It is measured as:

$$\Delta_{\text{EO}} = |\text{TPR}_{g_0} - \text{TPR}_{g_1}|. \quad (2.15)$$

A value of $\Delta_{\text{EO}} = 0$ indicates perfect equal opportunity; *lower is better*.

2.9.2 Equalized Odds

Equalized Odds [20] is a stricter criterion: the model must achieve equal TPR *and* equal TNR across groups simultaneously. This ensures both that qualified candidates are equally recognised and that unqualified ones are equally rejected:

$$\Delta_{\text{EOdd}} = \frac{1}{2} \left(|\text{TPR}_{g_0} - \text{TPR}_{g_1}| + |\text{TNR}_{g_0} - \text{TNR}_{g_1}| \right). \quad (2.16)$$

Again, *lower is better*. The relationship between these two metrics is: a model satisfying equalized odds also satisfies equal opportunity, but not vice versa.

2.9.3 Standard Accuracy

Standard accuracy is the fraction of correctly classified samples:

$$\text{Acc} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \quad (2.17)$$

While essential for gauging utility, accuracy alone is a misleading fairness proxy. On an imbalanced dataset biased towards one demographic group, a model can achieve high accuracy while systematically harming an underrepresented group.

2.9.4 Equalized Accuracy

To address the limitations of standard accuracy on skewed test distributions, the README paper [21] introduces *Equalized Accuracy*, which averages TPR and TNR across both groups equally:

$$\text{EAcc} = \frac{1}{4} \left(\text{TPR}_{g_0} + \text{TNR}_{g_0} + \text{TPR}_{g_1} + \text{TNR}_{g_1} \right). \quad (2.18)$$

Because each group contributes equally regardless of its size in the dataset, Equalized Accuracy is a more reliable indicator of a model’s generalisation across the population.

2.9.5 FPR-Based Fairness Metrics for Deepfake Detection

In the context of deepfake detection, where real faces vastly outnumber deepfakes in practice, *False Positive Rate* disparities can have severe societal consequences: real individuals being misclassified as deepfakes and subjected to unjust scrutiny [9]. Three metrics capture this:

1. **FPR Gap** (G_{FPR}): maximum pairwise difference in FPR across demographic groups:

$$G_{\text{FPR}} = \max_{g, g' \in \mathcal{G}} |\text{FPR}_g - \text{FPR}_{g'}|. \quad (2.19)$$

2. **Equal FPR** (F_{FPR}): summed absolute difference between each group’s FPR and the overall FPR:

$$F_{\text{FPR}} = \sum_{g \in \mathcal{G}} \left| \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1, G_i = g, Y_i = 0]}{\sum_i \mathbf{1}[G_i = g, Y_i = 0]} - \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1, Y_i = 0]}{\sum_i \mathbf{1}[Y_i = 0]} \right|. \quad (2.20)$$

3. **Fair Equalized Odds** (F_{EO}): extends Equal FPR to both positive and negative classes:

$$F_{\text{EO}} = \sum_{g \in \mathcal{G}} \sum_{q \in \{0,1\}} \left| \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1, G_i = g, Y_i = q]}{\sum_i \mathbf{1}[G_i = g, Y_i = q]} - \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1, Y_i = q]}{\sum_i \mathbf{1}[Y_i = q]} \right|. \quad (2.21)$$

For all three metrics, lower values indicate greater fairness.

2.9.6 Summary of Metrics

Table 2.1 summarises all evaluation metrics, their target conditions, and the direction of improvement.

2.10 Key Deep Learning Architectures Used in Fair AI

Table 2.2 provides a concise comparison of the core neural-network models referenced throughout this report.

Table 2.1: Summary of fairness and utility evaluation metrics used in this report. All fairness metrics are minimised; accuracy metrics are maximised.

Metric	Symbol	Condition checked	Better
Equal Opportunity	Δ_{EO}	TPR equal across groups	↓
Equalized Odds	Δ_{EOdd}	TPR & TNR equal	↓
Standard Accuracy	Acc	Overall correctness	↑
Equalized Accuracy	EAcc	Balanced group accuracy	↑
FPR Gap	G_{FPR}	Max FPR spread	↓
Equal FPR	F_{FPR}	FPR parity per group	↓
Fair Equalized Odds	F_{EO}	TPR & FPR parity	↓
AUC-ROC	AUC	Ranking quality	↑

Table 2.2: Key model architectures referenced in the fairness-in-AI literature covered in this report.

Model	Core idea	Strength	Limitation
VAE [14]	Probabilistic encoder-decoder with Gaussian latent prior	Smooth, generative latent space	No explicit control over which factors are captured
β -VAE [12]	Up-weight KL term by β	Encourages disentanglement	Hurts reconstruction at high β
FactorVAE [13]	Minimise Total Correlation via discriminator	Better disentanglement than β -VAE	Requires an additional discriminator
FFVAE [22]	Disentangle sensitive vs. non-sensitive latents	Fairness-focused VAE	Sensitive latents leak target attribute info
FD-VAE [21]	Three-way split: TAL, PAL, MAL + decorrelation loss	Explicitly decorrelates attributes	Requires target & protected attribute labels
Adversarial Autoencoder	GRL to strip demographic info from $\mathbf{z}$	Architecture-level fairness	Adversarial training instability
FAIR-FRFT (ours)	FrFT + adversarial autoencoder + CVaR	Frequency-domain fairness	Higher computational cost

2.11 Chapter Summary

This chapter introduced the mathematical and conceptual vocabulary needed to follow the technical discussions in subsequent chapters.

- **Mathematical tools:** KL divergence, Total Correlation, mutual information, ELBO, and CVaR provide the optimisation-theoretic backbone for fair representation learning.
- **Architectures:** Autoencoders, VAEs, and adversarial networks are the primary building blocks; disentanglement methods extend them to produce interpretable, attribute-separable latent spaces.
- **FrFT:** Offers a hybrid spatial–frequency domain well-suited to capturing deepfake artefacts that are less correlated with demographic features.
- **Evaluation metrics:** Equal Opportunity, Equalized Odds, Equalized Accuracy, and FPR-based metrics together give a multi-dimensional picture of fairness; no single metric suffices.

Chapter 3

README Framework: Fair Representation Learning Through Disentanglement

Declaration of Contributions:

- **Pradyuman Singh Shekhawat** — Reviewed key concepts including Variational Autoencoders, Total Correlation, and the disentanglement loss derivation; wrote Sections 3.1–3.4. Studied and explained the FD-VAE architecture, the decorrelation module, and the downstream classification network; wrote Sections 3.2–3.6.
 - **Vishesh Gupta** — Interpreted the fairness metrics and experimental results; wrote Sections 3.8.
-

3.1 Motivation and Problem Setting

One of the most natural entry-points for tackling bias in machine learning is to attack the problem at the level of *representation*: if the feature vector that a model learns contains no information about a sensitive attribute, then any classifier built on top of that vector cannot discriminate on the basis of that attribute. This idea is the backbone of the **README** (REpresentation learning by fairness-Aware Disentangling MEthod) paper by Park *et al.* [1].

Formally, let us denote the observed data by $\mathcal{X} = \{x_1, \dots, x_n\}$, the target attribute labels by $\mathcal{Y}_t = \{y_{t_1}, \dots, y_{t_n}\}$, and the protected attribute labels by $\mathcal{Y}_p = \{y_{p_1}, \dots, y_{p_n}\}$. A protected attribute is any characteristic such as *gender*, *age*, or *race* on which individuals

ought not to be discriminated against. A target attribute is the quantity the downstream classifier is actually supposed to predict, e.g., whether a face is *attractive* or whether the hair is *wavy*. The challenge is that in real-world datasets, target and protected attributes are often statistically correlated, for example, the “attractive” label in the CelebA dataset is influenced by the gender distribution in the data [1].

3.1.1 Limitation of Earlier Disentanglement Approaches

Prior to README, the state-of-the-art approach for fair representation learning via disentanglement was **FFVAE** (Flexibly Fair Variational AutoEncoder) by Creager *et al.* [23]. FFVAE partitions the latent space into two parts:

- *Sensitive latents*: latent variables trained to be highly informative about the protected attribute; and
- *Non-sensitive latents*: latent variables trained to be independent of the sensitive latents.

During downstream classification, only the non-sensitive latents are used, with the reasoning that this excludes all protected-attribute information from the prediction pipeline.

Park *et al.* identify a fundamental flaw in this two-way split. Because real-world attributes are correlated, the sensitive latents end up capturing *both* protected-attribute information *and* some target-attribute information. When these latents are excluded for fairness, useful discriminative signal for the target task is discarded along with the demographic information. The authors demonstrate this empirically: when they perform downstream classification using each subspace independently on the CelebA dataset (target attribute = *attractive*, protected attribute = *male*), the sensitive latents actually produce *higher* accuracy on the target attribute than the non-sensitive latents (67.4% vs 62.7%), revealing that excluding them comes at an unnecessary accuracy cost.

This observation motivates the central design choice of README: the latent space should be split into *three* independent subspaces, not two, separating target information, protected information, and the *shared* information predictive of both into its own dedicated slot.

3.2 Overview of the FD-VAE Architecture

The README framework is built around a model called the **Fairness-aware Disentangling Variational AutoEncoder (FD-VAE)**. At a high level, the FD-VAE is a VAE extended with two additional components: a discriminator for encouraging disentanglement across the three subspaces, and a decorrelation module that explicitly specifies *which type* of information should reside in each subspace.

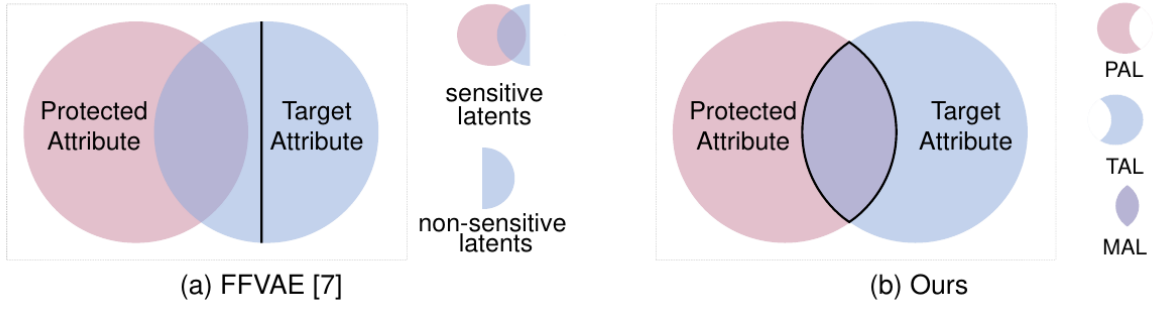


Figure 3.1: Motivation for fairness-aware disentanglement. (a) Prior methods (e.g., FFVAE) mix target and protected information, causing loss of useful features. (b) README separates them into TAL, PAL, and MAL, isolating shared information without harming performance.

The architecture operates in two stages:

1. **Representation Learning Stage:** The FD-VAE (encoder + decoder + discriminator + decorrelation module) is trained jointly to produce a well-disentangled and decorrelated latent space.
2. **Downstream Classification Stage:** The encoder is frozen, and a lightweight classification network is trained on the learned latent variables to predict the target attribute in a fairness-preserving manner.

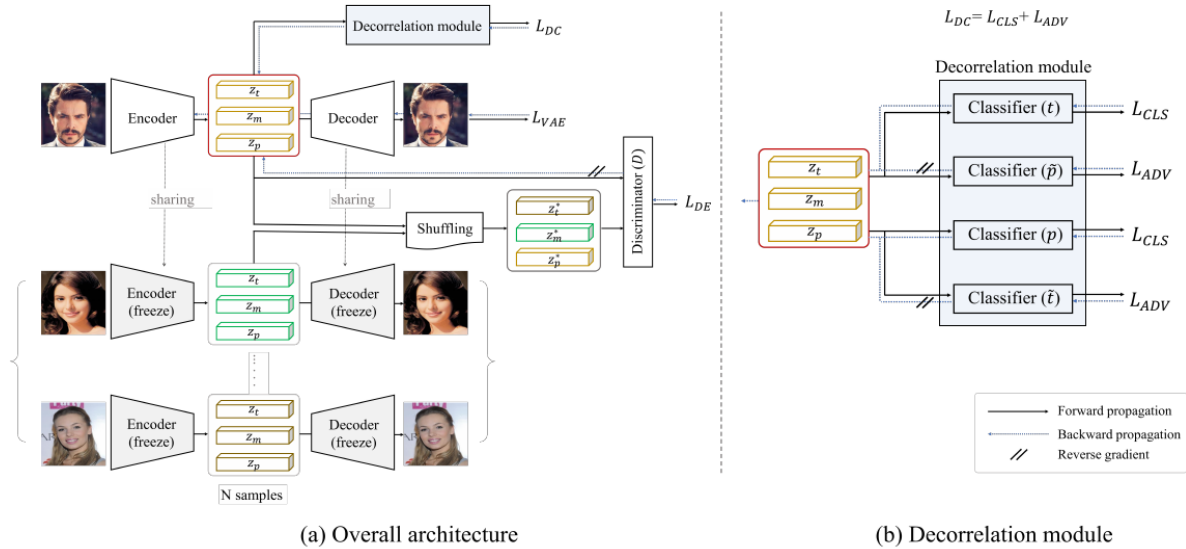


Figure 3.2: Full architecture of the Fairness-aware Disentangling VAE (FD-VAE). Source: adapted from Park *et al.* [1].

3.2.1 The Three Latent Subspaces

The encoder $q_{\Phi}(z | x)$ maps an input image to three independent latent subspaces:

Target Attribute Latent (TAL), z_t . This subspace is trained to carry only the information that is predictive of the target attribute y_t . During representation learning it receives a supervised classification signal for y_t and an adversarial signal that prevents it from leaking protected-attribute information.

Protected Attribute Latent (PAL), z_p . Symmetrically, this subspace is trained to contain only protected-attribute information y_p . Because it carries the demographic signal, it is *excluded entirely* from the downstream classification pipeline to ensure that the final predictor cannot use demographic information.

Mutual Attribute Latent (MAL), z_m . This is the key novelty of README. In real datasets there exists information that is predictive of *both* y_t and y_p simultaneously (e.g., facial structure features that correlate with both attractiveness and gender). If this shared information were forced into either TAL or PAL, it would violate the decorrelation objective. MAL provides a dedicated “overflow” subspace for this information. During downstream classification, the protected-attribute signal in MAL is stripped away before MAL’s remaining content is used to augment the classification.

In all experiments, each subspace is set to 20 dimensions, for a total latent dimension of 60.

3.3 The Variational Autoencoder Objective

The encoder and decoder follow the standard VAE formulation of Kingma and Welling [24]. The encoder outputs mean and variance parameters $(\mu, \sigma) = q_\Phi(x)$, from which latent codes are drawn via the reparameterisation trick:

$$z \sim \mathcal{N}(\mu_{q_\Phi}(x), \sigma_{q_\Phi}(x)).$$

The VAE objective is to maximise the Evidence Lower Bound (ELBO):

$$L_{\text{VAE}} = \sum_{i=1}^n \mathbb{E}_{q_\Phi(z_t, z_p, z_m | x_i)} [\log p_\Theta(x_i | z_t, z_p, z_m)] - \text{KL}[q_\Phi(z_t, z_p, z_m | x_i) \| p(z_t, z_p, z_m)]. \quad (3.1)$$

The first term is the *reconstruction loss*: the decoder p_Θ tries to faithfully reconstruct each input from the concatenated latent codes (z_t, z_p, z_m) . The second term is the *KL regularisation loss*: it encourages the posterior q_Φ to remain close to the standard Gaussian prior $p(z) = \mathcal{N}(0, I)$, which acts as a regulariser and ensures the latent space has good coverage.

3.4 Disentanglement Loss via Total Correlation

The VAE objective alone does not enforce that the three subspaces z_t , z_p , and z_m are *mutually independent*. Mutual independence between subspaces is formalised using the **Total Correlation (TC)** [25], which is the KL divergence between the joint distribution and the product of its marginals:

$$L_{\text{DE}} = \text{KL} \left[q_{\Phi}(z_t, z_p, z_m) \parallel \prod_{j \in S} q_{\Phi}(z_j) \right] = \mathbb{E}_{q_{\Phi}(z_t, z_p, z_m)} \left[\log \frac{q_{\Phi}(z_t, z_p, z_m)}{\prod_{j \in S} q_{\Phi}(z_j)} \right], \quad (3.2)$$

where $S = \{t, p, m\}$. Minimising the Total Correlation forces the three subspaces to become maximally independent of each other, which is the formal definition of disentanglement.

3.4.1 Adversarial Approximation of the Discriminator

Directly computing the log-density ratio in Equation (3.2) is intractable. Following the approach of Kim and Mnih [26], the log-density ratio is approximated by training a discriminator D to classify between *real samples* (drawn from the joint $q_{\Phi}(z_t, z_p, z_m)$) and *fake samples* $z^* = [z_t^*; z_p^*; z_m^*]$ generated by independently shuffling each subspace within a mini-batch:

$$L_{\text{DE}} \approx \mathbb{E}_{q_{\Phi}(z_t, z_p, z_m)} \left[\log \frac{D(z_t, z_p, z_m)}{1 - D(z_t, z_p, z_m)} \right]. \quad (3.3)$$

The discriminator is trained separately to classify real vs. fake by minimising:

$$L_D = -\mathbb{E}_{q_{\Phi}(z_t, z_p, z_m)} [\log D(z_t, z_p, z_m) + \log(1 - D(z_t^*, z_p^*, z_m^*))]. \quad (3.4)$$

The encoder q_{Φ} is simultaneously trained to fool the discriminator, which drives the Total Correlation toward zero and therefore encourages independence between the three subspaces.

3.5 Decorrelation Loss

The disentanglement loss ensures that the three subspaces are *statistically independent*, but it says nothing about *which information goes where*. The decorrelation loss L_{DC} takes over this role: it provides explicit supervision to ensure that z_t contains target attribute information, z_p contains protected attribute information, and each is purged of the other's signal.

The decorrelation loss is composed of two terms:

$$L_{\text{DC}} = \beta L_{\text{CLS}} + \gamma L_{\text{ADV}}, \quad (3.5)$$

where β and γ are hyperparameters (set to 5 and 10, respectively, in all experiments).

3.5.1 Classification Loss L_{CLS}

The classification loss provides direct supervision to assign the correct information to the correct subspace. It minimises cross-entropy for predicting the target label y_t from z_t , and for predicting the protected label y_p from z_p :

$$L_{\text{CLS}} = - \sum_{i=1}^n \mathbb{E}_{q_{\Phi}(z_t, z_p, z_m | x_i)} \left[\log p_t(y_{t_i} | z_t) + \log p_p(y_{p_i} | z_p) \right], \quad (3.6)$$

where p_t and p_p are lightweight classifiers (single fully-connected layers) trained to predict the target and protected attributes from their respective subspaces.

3.5.2 Adversarial Loss L_{ADV}

The classification loss alone is not enough: it pushes the correct information *into* each subspace but does not actively remove unwanted cross-information. The adversarial loss does this by training two adversarial classifiers $\tilde{p}$ (predicting y_p from z_t) and $\tilde{t}$ (predicting y_t from z_p), and using *reverse gradients* to force the encoder to make these classifiers fail:

$$L_{\text{ADV}} = \max_{\tilde{t}, \tilde{p}} \min_{\Phi} \sum_{i=1}^n \mathbb{E}_{q_{\Phi}(z_t, z_p, z_m | x_i)} \left[\log p_{\tilde{p}}(y_{p_i} | z_t) + \log p_{\tilde{t}}(y_{t_i} | z_p) \right]. \quad (3.7)$$

The minimax structure is crucial: the adversarial classifiers $\tilde{p}$ and $\tilde{t}$ are trained to *maximise* their ability to predict cross-attribute labels, while the encoder q_{Φ} is trained to *minimise* this ability. This creates a game that drives protected-attribute information out of z_t and target-attribute information out of z_p .

3.5.3 The Role of the Mutual Attribute Latent in Decorrelation

A key subtlety is what happens to information that is genuinely predictive of *both* attributes simultaneously. Forcing this information into either z_t or z_p would increase L_{ADV} in either direction. The MAL z_m acts as a pressure valve: the disentanglement loss L_{DE} indirectly encourages this shared information to migrate into z_m , because placing it there satisfies both the independence constraint and keeps L_{ADV} low. No additional explicit mapping is applied to z_m during representation learning.

3.5.4 Total Objective for Stage 1

The full representation learning objective that the FD-VAE maximises is:

$$L_{\text{TOTAL}} = L_{\text{VAE}} - \alpha L_{\text{DE}} - L_D - L_{\text{DC}}, \quad (3.8)$$

where α is a hyperparameter controlling the strength of the disentanglement objective ($\alpha = 50$ in all experiments). The negative signs on L_{DE} and L_D reflect the fact that the encoder is trained to *minimise* Total Correlation (thus maximise the negated discriminator objective), while the discriminator D is trained separately to minimise L_D .

3.6 Downstream Classification Network

Once Stage 1 training is complete, the encoder weights are frozen and the downstream classification network is trained to predict y_t in a fair manner.

Excluding PAL. The Protected Attribute Latent $\hat{z}_p$ is dropped entirely. Because the decorrelation objective has already separated all target-attribute information out of $\hat{z}_p$, this exclusion does not hurt target-prediction accuracy while guaranteeing that the final classifier has no access to demographic information.

Sanitising MAL. The Mutual Attribute Latent $\hat{z}_m$ still carries some residual protected-attribute signal (since it holds the shared information). A single fully connected layer f is applied to $\hat{z}_m$ to produce a transformed representation z_n . An adversarial protected-attribute classifier $\tilde{d}$ is trained to extract demographic information from z_n via reverse gradients, and f is trained to defeat $\tilde{d}$:

$$L_{\text{DCLS}} = \max_{d, f} \min_{\tilde{d}} \sum_{i=1}^n \mathbb{E}_{q_{\Phi}(\hat{z}_t, \hat{z}_p, \hat{z}_m | x_i)} \left[\log p_d(y_{t_i} | \hat{z}_t \oplus f(\hat{z}_m)) - \log p_{\tilde{d}}(y_{p_i} | f(\hat{z}_m)) \right], \quad (3.9)$$

where $\oplus$ denotes element-wise summation and d is the final target-attribute classifier.

Combining TAL and sanitised MAL. The sanitised representation z_n (from $f(\hat{z}_m)$) is added element-wise to $\hat{z}_t$ and the result is fed into the classifier d . This combination leverages the rich shared information captured in MAL (now stripped of its demographic component) to boost the discriminative power of the final prediction beyond what $\hat{z}_t$ alone could achieve.

Figure 3.3 illustrates the downstream classification architecture.

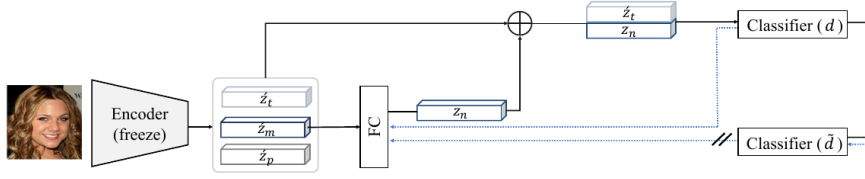


Figure 3.3: Downstream classification network of FD-VAE. Source: adapted from Park *et al.* [1].

3.7 Datasets

CelebA. The CelebA dataset [27] consists of approximately 200 000 celebrity face images, each annotated with 40 binary attributes. The authors consider four (protected, target) attribute pairs drawn from the attributes *male* (M), *young* (Y), *attractive* (A), *wavy hair* (W), and *big nose* (B):

$$(M, A), \quad (M, W), \quad (Y, A), \quad (Y, B).$$

UTK Face. The UTK Face dataset [28] contains face images spanning a wide age range, annotated for age, ethnicity, and gender. Gender is used as the protected attribute, while ethnicity (binarised into Caucasian vs. others) and age (binarised into young vs. older than 35) serve as target attributes. The training split is constructed to be *imbalanced* between the protected and target attributes, creating a realistic fairness challenge; the validation and test sets are balanced.

3.8 Experimental Results

Table 3.1 and Table 3.2 present the classification results on CelebA and UTK Face datasets, respectively, comparing FD-VAE against four baselines: a plain VAE (no fairness consideration), β -VAE, FactorVAE, and FFVAE. For each dataset the baseline methods are evaluated with the same 60-dimensional representation to ensure a fair comparison.

3.8.1 Results on CelebA

The results are striking. On the (M, A) pair, FD-VAE reduces the equal opportunity gap from FFVAE’s 17.7% to just **1.3%**, a reduction of 16.4 percentage points. Similarly, the equalized-odds gap falls from 17.5% to **4.9%**. Across all four attribute pairs on CelebA, FD-VAE consistently outperforms FFVAE by large margins on both fairness metrics while maintaining comparable accuracy (within 1–2% of FFVAE on equalized accuracy). This confirms that the three-way disentanglement preserves more discriminative signal for the target task than the simpler two-way split of FFVAE.

Table 3.1: Classification results on the CelebA dataset. M = male, Y = young, A = attractive, W = wavy hair, B = big nose. PA = protected attribute, TA = target attribute. ↓ indicates lower is better (fairer); ↑ indicates higher is better. Bold entries indicate the best result in each column. Source: Park *et al.* [1].

Method	TA	PA	Equal Opp. ↓	Eq. Odds ↓	Acc. ↑	Eq. Acc. ↑
VAE	A	M	27.2	28.7	70.6 / 65.3	,
β -VAE	A	M	21.1	22.1	68.7 / 64.6	,
FactorVAE	A	M	20.7	23.4	69.0 / 64.9	,
FFVAE	A	M	17.7	17.5	62.7 / 59.5	,
FD-VAE (Ours)	A	M	1.3	4.9	64.1 / 63.5	,
VAE	W	M	47.1	33.4	73.0 / 59.0	,
β -VAE	W	M	35.4	15.3	70.7 / 57.3	,
FactorVAE	W	M	38.8	28.8	71.0 / 58.0	,
FFVAE	W	M	21.1	15.4	67.7 / 54.3	,
FD-VAE (Ours)	W	M	13.4	8.9	66.1 / 52.9	,
VAE	A	Y	9.7	11.3	70.9 / 69.6	,
FFVAE	A	Y	8.0	8.2	64.2 / 62.9	,
FD-VAE (Ours)	A	Y	3.7	1.9	65.5 / 66.0	,
VAE	B	Y	17.0	19.2	67.2 / 64.0	,
FFVAE	B	Y	5.3	8.5	63.0 / 58.2	,
FD-VAE (Ours)	B	Y	1.2	1.5	63.9 / 57.7	,

3.8.2 Results on UTK Face

Table 3.2: Classification results on the UTK Face dataset. G = gender (protected), E = ethnicity (target), A = age (target). ↓ = lower is better; ↑ = higher is better. Bold = best result. Source: Park *et al.* [1].

Method	TA	PA	Equal Opp. ↓	Eq. Odds ↓	Acc. ↑
VAE	E	G	16.1	17.4	65.2
β -VAE	E	G	12.7	12.1	60.1
FactorVAE	E	G	12.1	12.9	59.4
FFVAE	E	G	9.8	9.1	59.7
FD-VAE	E	G	2.3	1.1	60.3
VAE	A	G	24.6	29.8	60.7
β -VAE	A	G	17.1	20.8	54.3
FactorVAE	A	G	23.2	27.1	54.6
FFVAE	A	G	13.6	17.2	54.5
FD-VAE	A	G	2.0	2.9	54.1

On UTK Face, FD-VAE achieves 2.3% and 2.0% equal opportunity for the ethnicity and age classification tasks respectively, reductions of roughly 7 to 11 percentage points over FFVAE with no meaningful drop in accuracy. This demonstrates that the improvement generalises across different datasets and protected-attribute types.

3.9 Summary

The README framework makes three distinct contributions to fair representation learning, which we briefly recap here before connecting them to the broader context of the report:

Three-way latent disentanglement. By introducing the Mutual Attribute Latent as a third subspace alongside the target and protected attribute latents, README avoids the information leak that afflicts two-way disentanglement approaches. This single design change enables a far cleaner separation of attribute-specific information.

Explicit decorrelation via adversarial supervision. The decorrelation loss goes beyond generic disentanglement: it provides direct supervision via L_{CLS} to assign correct information to each subspace, and adversarially removes unwanted cross-information via L_{ADV} . This two-pronged approach is shown in the ablation study to be the decisive factor for fairness.

Novel equalized accuracy metric. By proposing an accuracy metric that weights all group–outcome combinations equally, README provides the community with a tool that is immune to dataset imbalance, making comparisons across skewed benchmarks more meaningful.

The experimental results on CelebA and UTK Face demonstrate that FD-VAE outperforms all prior methods by substantial margins on standard fairness metrics (equal opportunity, equalized odds), while achieving comparable accuracy. The framework lays the groundwork for the application we explore in Chapter 4: extending fairness-aware representation learning to the more challenging problem of deepfake detection.

The primary reference for this chapter is Park et al., “README: REpresentation learning by fairness-Aware Disentangling MEthod,” arXiv:2007.03775v1, 2020. All tables and figures are adapted from this paper.

Chapter 4

CVaR-Based Fair Deepfake Detection: DAG-FDD and DAW-FDD

Declaration of Contributions: This chapter was primarily authored by *Aryan Choudhary*, who conducted a detailed study of the referenced work, systematically extracted its mathematical formulations and reformulated the methodology into this report chapter. The presentation of CVaR-based robust optimization along with demographic-agnostic and demographic-aware fairness frameworks was carefully structured and formalized to align with the objectives of the literature review. All mathematical expressions and derivations were rigorously verified to ensure consistency with the original paper.

4.1 Introduction

In this chapter, we review the fairness oriented deepfake detection framework proposed in [2], with particular emphasis on the role of Conditional Value-at-Risk (CVaR) in robust optimisation. The central idea is that conventional training, which minimises average loss, tends to prioritise easy and dominant samples. In deepfake detection, this can lead to uneven performance across hidden or explicit demographic groups: a model may achieve strong overall accuracy while still producing substantially higher error rates for certain subpopulations.

To address this limitation, the paper introduces two related training strategies:

- **DAG-FDD** (Demographic-Agnostic Fair Deepfake Detection) : used when demographic labels are *not* available.

- **DAW-FDD** (Demographic-Aware Fair Deepfake Detection) : used when demographic labels *are* available.

Both methods are built on the same broad principle: instead of optimising the average loss over all samples, the training procedure is modified to emphasise the worst-performing samples or groups. This makes the detector more robust and in the fairness setting, reduces the likelihood that the model performs well only on majority or easy cases.

The contribution of this paper [2] is particularly significant as it demonstrates that fairness can be incorporated through the objective function without requiring changes to the underlying detection architecture. Since the fairness behavior is largely governed by the loss formulation, the approach remains flexible and can be extended to a wide range of models.

4.2 Background: Why Average Loss Is Not Enough

Standard learning methods minimise the expected loss

$$R_{\text{avg}}(\theta) = \mathbb{E}_{(X,Y) \sim P} [\ell(\theta; X, Y)], \quad (4.1)$$

where θ denotes the model parameters, (X, Y) is a training sample drawn from data distribution P , and $\ell(\theta; X, Y)$ is the prediction loss. This objective treats every sample equally on average. While effective for ordinary accuracy maximisation, it is not suitable when the dataset contains imbalance across subgroups: a model can reduce its total average loss by doing well on dominant or easier samples, even if it performs poorly on minority or difficult groups.

This is exactly the limitation which we try to overcome. The authors of [2] argue that fairness in deepfake detection should not be measured only through average behaviour, because the average can hide worst-case performance. A detector that fails badly on one group but performs well on others may still appear strong under standard metrics. Therefore, the paper replaces average-risk training with a *robust risk* objective.

4.3 CVaR as the Core Robustness Principle

The paper [2] uses **Conditional Value-at-Risk (CVaR)** as the main robust optimisation tool. CVaR is a classical idea from risk-sensitive optimisation; it focuses on the *tail* of the loss distribution rather than its mean. For a given parameter vector θ , the CVaR objective is

$$\text{CVaR}_\alpha(\theta) = \inf_{\lambda \in \mathbb{R}} \left\{ \lambda + \frac{1}{\alpha} \mathbb{E}_{(X,Y) \sim P} [[\ell(\theta; X, Y) - \lambda]_+] \right\} \quad (4.2)$$

The notion of infimum has been formally introduced in Chapter 2.

where $[a]_+ = \max(0, a)$.

4.3.1 Interpretation of the variables

Here, λ is a threshold or pivot value. The expression $[\ell(\theta; X, Y) - \lambda]_+$ retains only the part of the loss that exceeds λ : samples whose losses are below the threshold contribute nothing, while samples above the threshold contribute proportionally to how badly the model performs on them.

The parameter $\alpha \in (0, 1)$ determines the fraction of the highest-loss region that is emphasised. A smaller α concentrates more strongly on the hardest samples, while a larger α makes the objective closer to average loss. This makes α an important hyperparameter in controlling how *aggressively* the model focuses on the difficult portion of the data.

4.3.2 Intuition

The intuition behind CVaR is simple but powerful: instead of asking “How well does the model perform on average?”, it asks “How bad is the model on the worst fraction of cases?” This is especially useful in fairness-sensitive applications, because the worst fraction of cases is often where minority or underrepresented groups appear.

4.4 DAG-FDD: Demographic-Agnostic Fair Deepfake Detection

4.4.1 Motivation

DAG-FDD is used when demographic labels are not available which is a realistic scenario in many deepfake detection settings, since training sets may not contain explicit annotations for race, sex, age or other sensitive attributes. Even without demographic labels, the data may still contain *latent groups*, some of which may be harder to classify than others. DAG-FDD is designed to reduce the risk that the detector performs disproportionately poorly on such hidden subgroups.

4.4.2 Latent-Group Formulation

The paper models the overall data distribution as a mixture of K latent group distributions:

$$P = \sum_{m=1}^K \pi_m P_m \quad (4.3)$$

where P_m is the distribution of the m -th latent group, π_m is its mixture weight, and $\sum_m \pi_m = 1$. The corresponding worst-group risk is

$$R_{\max}(\theta) = \max_{m=1,\dots,K} \mathbb{E}_{(X,Y) \sim P_m} [\ell(\theta; X, Y)] \quad (4.4)$$

This objective directly represents the fairness concern: instead of minimising the mean risk across all groups, we control the risk of the worst group.

4.4.3 Why CVaR Helps in the Demographic-Agnostic Setting

The paper [2] shows that CVaR on sample losses can act as an upper-bound surrogate for worst-group risk when α is small enough. Specifically, if

$$\alpha \leq \min_m \pi_m \quad (4.5)$$

then the CVaR objective provides an upper bound on the maximum group risk. This result is important because it justifies using sample-level CVaR even when group labels are missing. The practical meaning is that the hardest samples in the overall dataset tend to come from the hardest latent groups; therefore, if training is forced to pay greater attention to the worst samples, the model indirectly becomes more robust to hidden demographic imbalance.

4.4.4 Empirical Objective Used in DAG-FDD

For a dataset of size n , the empirical DAG-FDD objective is

$$\mathcal{L}_{\text{DAG-FDD}}(\theta, \lambda) = \lambda + \frac{1}{\alpha n} \sum_{i=1}^n [\ell(\theta; X_i, Y_i) - \lambda]_+ \quad (4.6)$$

This is the sample-based approximation of the population CVaR objective (4.2). The term λ sets the threshold level; the summation considers only samples whose loss exceeds the threshold; and the factor $1/(\alpha n)$ rescales the tail contribution so that the hardest samples dominate the update. In effect, DAG-FDD behaves like a principled version of hard-example mining.

4.4.5 Optimisation of DAG-FDD

The paper optimises DAG-FDD by alternating between the threshold parameter λ and the model parameters θ . For fixed θ , the objective is convex in λ , so λ can be found efficiently by binary search. For fixed λ , the model parameters are updated by gradient descent:

$$\theta^{l+1} = \theta^l - \frac{\eta}{\alpha |S_b|} \sum_{i \in S_b} \frac{\partial}{\partial \theta} \ell(\theta^l; X_i, Y_i) \cdot \mathbf{1}[\ell(\theta^l; X_i, Y_i) > \lambda] \quad (4.7)$$

where S_b is the mini-batch, η is the learning rate, and $\mathbf{1}[\cdot]$ is the indicator function. Only the samples with losses above the threshold contribute to the gradient.

It is the main mechanism by which DAG-FDD shifts learning towards difficult samples.

4.4.6 Interpretation

The demographic-agnostic setting is especially useful because it does not require any sensitive labels, making the method broadly applicable. The fairness guarantee is indirect: the method cannot guarantee equality across explicit groups because those groups are not known. Instead, it controls the worst-case behaviour among latent subpopulations, still very valuable, since hidden demographic imbalance is common in real datasets.

4.5 DAW-FDD: Demographic-Aware Fair Deepfake Detection

4.5.1 Motivation

DAW-FDD is used when demographic labels are available (e.g. race or sex labels). In this case, fairness can be enforced more explicitly and directly. Unlike DAG-FDD, which treats the data as a collection of unlabelled samples, DAW-FDD recognises the existence of demographic groups and explicitly tries to reduce performance disparity among them.

4.5.2 Group Risk

Let $\mathcal{G}$ denote the set of demographic groups. For a sample i belonging to group g , the group-specific risk is

$$R_g(\theta) = \mathbb{E}_{(X,Y)|G=g}[\ell(\theta; X, Y)] \quad (4.8)$$

The fairness goal is to avoid situations where some groups experience substantially larger risk than others, particularly important in deepfake detection, since a detector that is less accurate for one demographic group may amplify unfairness in downstream applications.

4.5.3 Group-Level CVaR

The paper also introduces a CVaR objective over group risks [2]:

$$\text{CVaR}_\alpha^{\mathcal{G}}(\theta) = \inf_{\lambda \in \mathbb{R}} \left\{ \lambda + \frac{1}{\alpha} \mathbb{E}_{G \sim \text{Uniform}(\mathcal{G})} [[R_g(\theta) - \lambda]_+] \right\} \quad (4.9)$$

This differs from DAG-FDD in an important way: the quantity being evaluated is not the loss of individual samples, but the loss of entire groups. The method therefore places direct pressure on the worst-performing demographic groups. The paper further notes that the group-CVaR objective can be interpreted as balancing the average risk across groups and the deviation from the worst groups, so DAW-FDD actively suppresses disparity rather than merely optimising accuracy under group awareness.

4.5.4 Per-Group Top k / CVaR Estimation

Why an Additional Mechanism Is Needed

One major issue in the demographic-aware setting is that each group may itself contain an imbalanced class composition. For example, a group may have many more real samples than fake samples, or vice versa. If one simply averages losses within each group, the majority class may dominate the group-risk estimate. To address this, the paper introduces a top- k (or CVaR-style) estimator *inside* each group [2], ensuring that the hardest examples within a group contribute more strongly to the group loss.

Group-Level Top k Loss

For a group g , the group loss is computed in a CVaR-equivalent form over the hardest fraction of its samples:

$$\mathcal{L}_g(\theta) = \min_{\lambda_g \in \mathbb{R}} \left\{ \lambda_g + \frac{1}{\alpha_g n_g} \sum_{i \in \mathcal{I}_g} [\ell(\theta; X_i, Y_i) - \lambda_g]_+ \right\} \quad (4.10)$$

where $\mathcal{I}_g$ is the index set of samples in group g , $n_g = |\mathcal{I}_g|$, α_g controls how much of the group's tail is retained and λ_g is a group-specific threshold.

Role of α_g

The parameter α_g determines the degree of emphasis on the hardest samples within a group: a smaller value means only the worst examples are considered, while a larger value includes a broader portion of the group. In practice, α_g prevents a group's loss estimate from being diluted by easy or majority-class samples, especially important in deepfake detection, where the real/fake imbalance within each group can otherwise distort the fairness objective.

4.5.5 Final DAW-FDD Objective

The overall DAW-FDD objective is *hierarchical*. It first computes a robust loss within each group and then applies another CVaR layer over the groups themselves:

$$\mathcal{L}_{\text{DAW-FDD}}(\theta, \lambda) = \lambda + \frac{1}{\alpha |\mathcal{G}|} \sum_{g \in \mathcal{G}} [\mathcal{L}_g(\theta) - \lambda]_+ \quad (4.11)$$

where $\mathcal{L}_g(\theta)$ is given by Equation (4.10).

How to Interpret the Two Levels

This objective embeds two nested robustness mechanisms:

- **Inner level:** within each demographic group, focus on the hardest samples.
- **Outer level:** across all groups, focus on the worst-performing groups.

DAW-FDD is therefore more structured than DAG-FDD. It does not merely emphasise hard samples globally, but explicitly separates the fairness problem into two levels: sample-level imbalance and group-level disparity.

Optimisation Procedure

The paper optimises DAW-FDD by alternating among three types of variables: the group thresholds λ_g , the global threshold λ , and the model parameters θ . *A sample contributes to the update only if it is hard relative to its group, and a group contributes strongly only if it is hard relative to the overall population.* This double filtering effect encourages fairness both *within* and *across* groups.

4.6 Relationship Between DAG-FDD and DAW-FDD

The two methods are closely related, but solve slightly different versions of the same fairness problem. Table 4.1 summarises the key differences.

Table 4.1: Comparison of DAG-FDD and DAW-FDD.

Property	DAG-FDD	DAW-FDD
Demographic labels required	No	Yes
Group structure assumed	Latent (implicit)	Explicit
CVaR applied to	Individual samples	Groups + samples
Fairness guarantee	Worst latent group	Worst explicit group
Complexity	Lower	Higher (hierarchical)

In summary, DAG-FDD can be viewed as a *latent-group* robust optimisation method, whereas DAW-FDD is a *hierarchical group-robust* optimisation method. The former protects against hidden worst-case behaviour; the latter protects against known demographic disparities. From a fairness perspective, DAW-FDD is more informative when group annotations are available, but DAG-FDD is more practical when such annotations are missing.

4.7 Experimental Results Reported in the Paper

The experimental setup in [2] is designed to evaluate whether CVaR-based objectives can improve fairness without degrading detection performance. The evaluation spans multiple datasets, architectures, and fairness settings.

Experimental Setup Overview

- **Datasets:** FaceForensics++ (FF++), Celeb-DF, DeepFakeDetection (DFD), and DFDC.
- **Backbones:** Xception, ResNet-50, EfficientNet-B3, DSP-FWA, and RECCE.
- **Fairness Metrics:** GFPR, FFPR, FEO (measuring group-wise false-positive disparities across gender, race and intersection).
- **Detection Metrics:** AUC, FPR, TPR, ACC

A key design choice in the paper is the emphasis on **false-positive fairness**, since misclassifying real content as fake can have significant real-world consequences.

4.7.1 Key Observations

The results across all experiments exhibit several consistent trends:

- **Fairness improvement is consistent:** Both DAG-FDD and DAW-FDD reduce disparity in GFPR, FFPR and FEO across all datasets.
- **No strong fairness–accuracy trade-off:** In multiple cases, detection performance (AUC, ACC) is preserved or even improved.
- **DAG vs DAW behaviour:**
 - DAG-FDD performs well without demographic labels by focusing on hard samples.
 - DAW-FDD performs better when labels are available due to explicit group-level optimisation.

- **Intersectional fairness improves:** Gains are not limited to single attributes; intersection groups (race + gender) also show reduced disparity.

4.7.2 Results on FF++ (In-Domain Analysis)

Methods	Require Demographics	Fairness Metrics (%) ↓									Detection Metrics (%)			
		Gender			Race			Intersection			Overall			
		G_{FPR}	F_{FPR}	F_{EO}	G_{FPR}	F_{FPR}	F_{EO}	G_{FPR}	F_{FPR}	F_{EO}	AUC ↑	FPR ↓	TPR ↑	ACC ↑
Original	—	4.10	4.10	9.06	13.09	17.28	21.00	17.93	31.59	53.95	92.76	22.06	94.43	91.49
DRO _{χ²} [30]	×	2.68	2.68	6.75	8.32	8.97	20.40	8.73	22.97	55.54	97.18	6.32	90.25	90.86
DAG-FDD (Ours)		1.63	1.63	6.21	8.23	9.53	11.49	9.65	21.21	48.10	97.13	9.54	94.32	93.63
Naive [16]	✓	11.98	11.98	18.20	16.57	22.01	25.97	28.90	72.19	93.59	83.17	50.77	92.62	84.87
FRM [22]		1.33	1.33	5.88	9.24	12.75	20.13	10.39	25.57	60.90	97.81	4.76	90.85	91.63
Group DRO [59]		8.20	8.20	12.87	14.37	20.97	23.79	21.86	44.98	65.24	91.13	27.83	95.15	91.04
Cons. EFPR [60]		4.24	4.24	7.91	7.09	7.49	12.41	14.95	27.80	46.62	94.30	22.61	94.94	91.80
Cons. EO [60]		1.77	1.77	4.79	10.92	12.61	16.50	17.25	26.95	44.68	95.74	16.28	95.89	93.72
DAW-FDD (Ours)		0.32	0.32	3.99	2.49	3.88	6.29	6.61	14.06	33.84	97.46	11.46	95.40	94.17

Figure 4.1: Comparison of fairness-aware methods on FF++ using the Xception backbone (adapted from [2]).

Datasets	Methods	Fairness Metrics (%) ↓									Detection Metrics (%)			
		Gender			Race			Intersection			Overall			
		G_{FPR}	F_{FPR}	F_{EO}	G_{FPR}	F_{FPR}	F_{EO}	G_{FPR}	F_{FPR}	F_{EO}	AUC ↑	FPR ↓	TPR ↑	ACC ↑
Celeb-DF	Original	4.93	4.93	22.04	3.31	4.77	26.06	11.81	15.66	39.95	97.17	13.01	95.83	94.05
	DAG-FDD (Ours)	2.02	2.02	16.77	1.20	1.22	28.56	2.54	3.09	30.43	98.00	2.42	87.40	89.44
	DAW-FDD (Ours)	3.81	3.81	18.93	3.14	3.34	33.91	3.80	4.91	35.48	98.03	2.10	84.53	87.21
DFD	Original	2.95	2.95	5.52	7.35	7.35	7.72	8.67	15.81	24.31	92.94	25.00	96.01	89.09
	DAG-FDD (Ours)	2.92	2.92	4.79	6.08	6.08	7.05	8.30	13.52	19.57	93.40	28.07	96.31	88.28
	DAW-FDD (Ours)	1.40	1.40	3.14	2.36	2.36	3.35	7.20	8.74	14.70	93.17	27.75	95.95	88.14
DFDC	Original	1.64	1.64	4.36	4.02	5.85	38.84	20.17	38.68	119.71	92.40	7.28	76.32	86.87
	DAG-FDD (Ours)	1.30	1.30	5.38	4.50	5.78	46.56	14.65	33.79	113.93	92.69	6.61	74.41	86.61
	DAW-FDD (Ours)	1.73	1.73	3.39	3.48	4.14	42.87	11.19	23.63	115.15	94.88	4.27	75.10	88.37

Figure 4.2: Evaluation on Celeb-DF, DFD, and DFDC using Xception (adapted from [2]). Results show fairness and detection metrics across datasets.

From Figure 4.1, the following patterns can be observed:

- **Strong reduction in fairness gaps:** Both DAG-FDD and DAW-FDD significantly lower GFPR/FFPR compared to the original model.
- **DAW-FDD achieves best fairness:** When demographic labels are available, DAW-FDD consistently produces the lowest group disparities.
- **Competitive detection performance:** Improvements in fairness do not come at the cost of AUC or accuracy.
- **Comparison with prior fairness methods:** The proposed methods outperform baselines such as DRO, Group DRO, and constraint-based approaches.

This confirms that CVaR-based optimisation is effective even in standard in-domain training.

4.7.3 Cross-Dataset Generalisation

Figure 4.2 evaluates generalisation across datasets.

- **Fairness improvements persist across datasets:** Both DAG-FDD and DAW-FDD continue to reduce disparity outside the training dataset.
- **Robustness to distribution shift:** The method remains effective even when data characteristics change (e.g., DFDC vs FF++).
- **DAW-FDD benefits from group labels:** When demographic structure is available, improvements are more pronounced.
- **DAG-FDD remains practical:** Even without labels, DAG-FDD provides meaningful robustness gains.

This indicates that the CVaR objective is not overfitting to a specific dataset distribution.

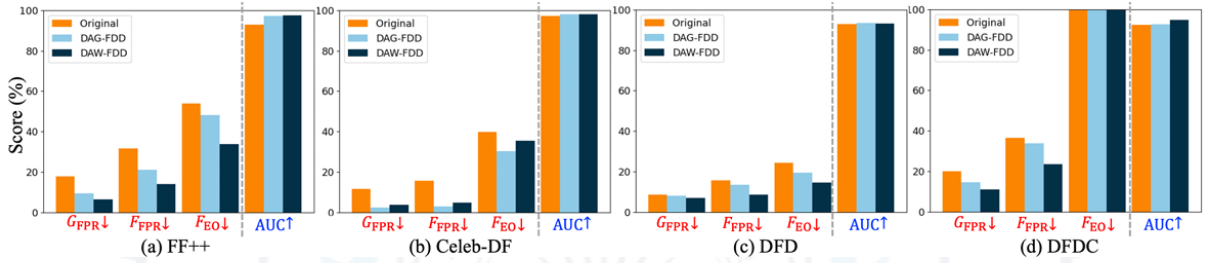


Figure 4.3: Intersectional fairness comparison of the Xception detector across four datasets: FF++, Celeb-DF, DFD, and DFDC (adapted from Figure 1 in [2]).

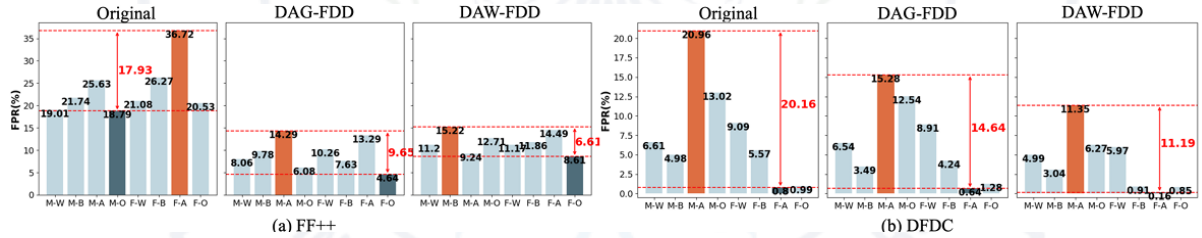


Figure 4.4: FPR comparison of the Xception detector on intersectional groups for FF++ and DFDC (adapted from Figure 2 in [2]). Orange and dark cyan bars denote the groups with the highest and lowest FPR, respectively, and the red bracket shows the gap, where a smaller value is better.

Figure 4.3 provides a more detailed view of the intersectional setting across datasets. It shows that the proposed methods consistently *reduce the fairness gaps for intersection groups while maintaining competitive AUC*. The improvement is visible not only on FF++, but also on Celeb-DF, DFD, and DFDC, which supports the paper’s claim that the CVaR-based objectives generalize well beyond a single benchmark.

Figure 4.4 focuses specifically on false-positive disparity across intersection groups. This is especially important because the paper emphasizes FPR as the most practically harmful error mode in deepfake detection. Both DAG-FDD and DAW-FDD reduce the gap between the highest and lowest FPR groups, showing that the fairness objective improves not just average behavior but also worst-group outcomes.

4.7.4 Backbone Robustness

Models	Methods	Fairness Metrics (%) ↓									Detection Metrics (%)			
		Gender			Race			Intersection			Overall			
		G_{FPR}	F_{FPR}	F_{EO}	G_{FPR}	F_{FPR}	F_{EO}	G_{FPR}	F_{FPR}	F_{EO}	AUC ↑	FPR ↓	TPR ↑	ACC ↑
ResNet-50	Original	2.58	2.58	6.64	12.80	15.45	17.62	20.06	43.18	60.66	94.32	25.96	96.36	92.37
	DAG-FDD (Ours)	2.21	2.21	6.37	6.44	11.42	14.17	16.68	37.50	52.79	94.51	22.52	95.34	92.15
	DAW-FDD (Ours)	3.78	3.78	10.21	7.01	8.27	13.09	13.28	35.08	60.07	93.70	23.56	93.65	90.58
EfficientNet-B3	Original	1.97	1.97	4.15	9.05	10.86	14.12	13.38	22.65	40.13	95.91	20.25	97.21	94.09
	DAG-FDD (Ours)	0.47	0.47	5.36	9.48	9.58	13.50	10.87	19.34	46.08	97.20	8.40	92.87	92.65
	DAW-FDD (Ours)	0.04	0.04	5.53	3.79	4.67	12.63	6.43	12.57	43.72	96.30	8.22	91.43	91.49
DSP-FWA	Original	5.90	5.90	11.81	11.07	14.58	21.98	21.38	48.20	75.91	91.79	31.64	93.17	88.74
	DAG-FDD (Ours)	4.64	4.64	9.77	12.52	18.04	25.03	15.61	40.57	74.54	91.47	32.35	93.70	89.05
	DAW-FDD (Ours)	3.02	3.02	11.30	5.75	10.52	19.34	12.84	36.05	75.73	90.84	30.43	91.97	87.97
RECCE	Original	0.87	0.87	3.14	18.81	27.65	30.07	30.26	67.38	80.34	98.05	21.20	98.21	94.74
	DAG-FDD (Ours)	0.55	0.55	3.71	12.68	17.41	20.33	15.40	36.17	54.24	98.33	12.01	96.80	95.23
	DAW-FDD (Ours)	0.25	0.25	4.75	6.99	7.96	11.95	13.54	23.44	52.95	98.35	8.15	94.59	94.10

Figure 4.5: Performance across multiple backbones (ResNet-50, EfficientNet-B3, DSP-FWA, RECCE) on FF++ (adapted from [2]).

Figure 4.5 demonstrates that the proposed method is not tied to a specific architecture.

- **Backbone-agnostic behaviour:** Fairness improvements are consistent across all evaluated detectors.
- **Stable detection metrics:** Performance remains competitive regardless of architecture.
- **Plug-and-play nature:** The method only modifies the loss function and can be integrated into existing models.

4.7.5 Computational Considerations

- **Additional overhead:** Comes from optimisation of CVaR thresholds (λ , λ_g).
- **Implementation cost:** Requires binary search or iterative updates.
- **Practical trade-off:** The overhead is modest relative to the fairness gains achieved.

4.8 Key Insights for Our Project

This paper is useful for our project because it offers a clean way to promote fairness without redesigning the full architecture. The main lesson we draw from it is that **fairness can be controlled at the optimisation level**. The most important takeaways are as follows.

1. Worst-case focus is more informative than average loss when the goal is fair and robust performance.
2. Hierarchical CVaR is a natural way to handle fairness when both sample-level hardness and group-level imbalance are present.

3. DAG-FDD is the appropriate formulation when demographic labels are not available.
4. DAW-FDD is the more structured and more explicit formulation when demographic labels are available.
5. The top- k or CVaR estimator inside each group is important for handling imbalance between real and fake samples within a demographic group.

4.9 Conclusion

In this chapter, we described the CVaR-based fairness framework for deepfake detection and explained how it leads to two related methods: DAG-FDD and DAW-FDD. DAG-FDD handles the setting where demographic labels are absent and works through latent worst-case sample optimisation. DAW-FDD extends the same idea to the demographic-aware setting by adding group-level CVaR and per-group top- k estimation.

The main contribution of the paper is that it replaces average-risk training with a more robust and fairness-sensitive objective, allowing the detector to pay greater attention to the hardest samples and the worst-performing groups, which in turn improves fairness without significantly compromising detection quality. For our own work, this paper serves as a strong reference for using robust optimisation as a fairness mechanism in deepfake detection.

Chapter 5

Targeting a Critical Gap in Deepfake Detectors

Declaration of Contributions: This chapter was developed by *Siddharth Gupta* and *Aryan Choudhary*, the former contributed to the conceptualisation and theoretical design of the autoencoder architecture and the debiasing framework, based on an extensive review of existing literature, while the later contributed to the theoretical formulation and design of the CVaR-based fairness module.

This chapter presents **FAIR-FRFT** (*Fairness-Aware Fractional Fourier Transform for Deepfake Detection*), our proposed framework for demographically fair deepfake detection. Unlike prior methods that append fairness as a post-hoc constraint on top of an existing architecture, FAIR-FRFT treats fairness as a *fundamental design principle*, embedding demographic invariance directly into the representation-learning process through three tightly coupled components: (i) a hybrid spatial–frequency encoder driven by the Fractional Fourier Transform, (ii) an adversarial autoencoder with a Gradient Reversal Layer, and (iii) a CVaR-based tail-risk fairness objective. Together, these components ensure that the learned latent representations are simultaneously deepfake-discriminative and demographic-agnostic.

5.1 Problem Statement

Deepfake detectors are not necessarily fair across demographic groups. A model may achieve strong overall accuracy while still producing systematically different error rates for different communities. This occurs when the detector learns correlations present in the training data that do not generalise fairly across groups.

In the context of deepfake detection, such bias is especially problematic because the impact of an error can be severe. A detector that performs well on one community but poorly on another may create uneven trust and unequal risk.

Illustrative Scenario

Consider an election setting in which a deepfake speech is circulated publicly against a political candidate. Suppose the detector, due to its inherent bias and limitations in learned representations, incorrectly labels the speech as genuine rather than synthetic. Such a failure may allow the manipulated content to spread widely, potentially triggering public unrest, misleading voters and damaging the candidate’s reputation. This example highlights why fairness in deepfake detection is not merely a technical concern, but a *matter of real-world safety and social responsibility*.

5.2 Overview of the FAIR-FRFT Architecture

Figure 5.1 provides a complete view of the FAIR-FRFT pipeline. At the highest level, the framework comprises four functional blocks:

1. **FrFT pre-processor** The input image $\mathbf{x}$ is mapped to the hybrid spatial frequency domain via the Fractional Fourier Transform, yielding frequency maps $\mathbf{x}' = \mathcal{F}_\alpha^{(d)}(\mathbf{x})$.
2. **Adversarial autoencoder** An encoder E_θ compresses $\mathbf{x}'$ into a latent representation $\mathbf{z}$; a decoder G_θ reconstructs the spatial-domain image $\hat{\mathbf{x}}$ from $\mathbf{z}$. The cycle is closed by re-encoding the reconstruction for latent consistency.
3. **Demographic adversary** A demographic classifier D_ϕ attempts to predict the protected group g from $\mathbf{z}$. Gradient reversal forces the encoder to *maximise* the adversary’s error, thereby stripping demographic information from the latent space.
4. **Binary Real/Fake classifier** After the encoder has converged to a fair representation, a lightweight classifier C_ψ is trained on the frozen $\mathbf{z}$ to detect real versus fake faces.

5.3 Fractional Fourier Transform Pre-processing

The Fractional Fourier Transform (FrFT), introduced in Section 2.7, serves as the input representation for FAIR-FRFT. For an order parameter $\alpha \in [0, 1]$, the 2-D FrFT of an image $\mathbf{x}$ is computed channel-wise as:

$$\mathbf{x}' = \mathcal{F}_\alpha^{(d)}(\mathbf{x}) \quad (5.1)$$

$$\mathcal{F}_\alpha[f](u) = e^{-i\pi u^2 \cot \alpha} \cdot \text{FFT} \left[f(t) e^{-i\pi t^2 \cot \alpha} \right] (u \csc \alpha) \quad (5.2)$$

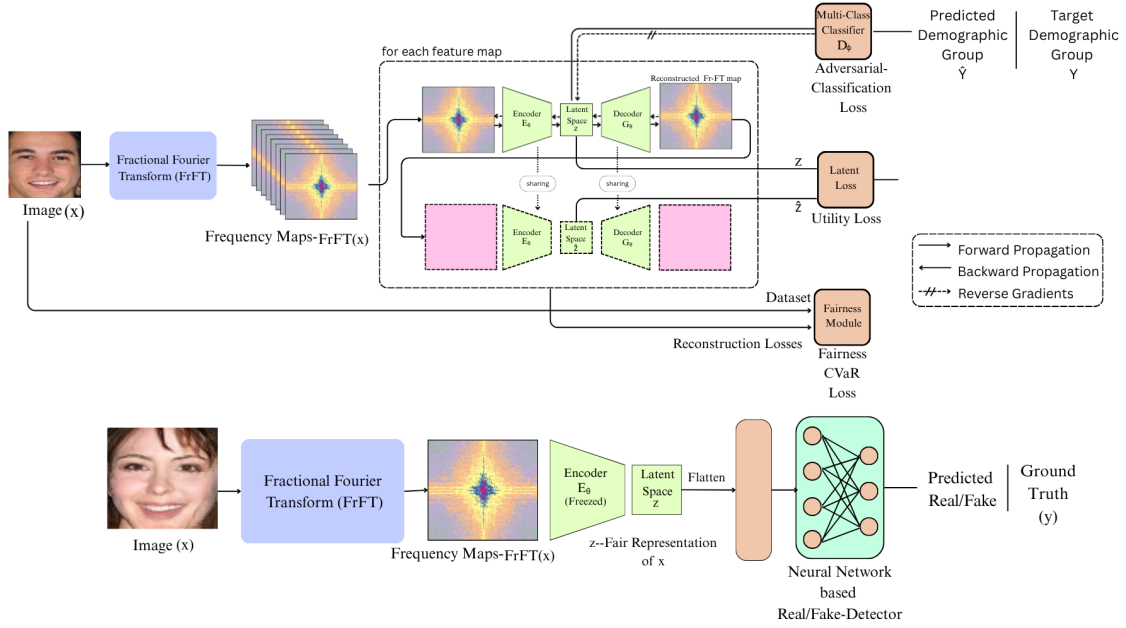


Figure 5.1: FAIR-FRFT Architecture. Top: FrFT-transformed input x' is encoded by E_θ into latent z , reconstructed by G_θ to $\hat{x}$, and re-encoded for consistency. The model includes demographic adversary D_ϕ , real/fake head R_ψ , and CVaR fairness loss. Bottom: After training, the encoder is frozen and a classifier predicts real/fake labels from the learned fair representation.

5.4 Adversarial Autoencoder

Encoder and Decoder

The encoder $E_\theta : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^d$ maps the FrFT-transformed image to a d -dimensional latent vector:

$$\mathbf{z} = E_\theta(\mathbf{x}'). \quad (5.3)$$

We implement E_θ as a three-layer CNN with filter counts 32, 64 and 128. Layer Normalization after each convolution and additive Gaussian noise ($\sigma = 0.05$) for regularisation. Critically, no non-linear activation function is applied after the final layer; this preserves the sign of latent activations, which carries phase information propagated from the FrFT.

The decoder $G_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^{H \times W \times C}$ reconstructs the spatial-domain image:

$$\hat{\mathbf{x}} = G_\theta(\mathbf{z}). \quad (5.4)$$

G_θ is a symmetric CNN with transposed convolutions and skip connections from the

corresponding encoder layers, mirroring the U-Net paradigm to recover fine spatial detail. The reconstructed image is then re-processed through FrFT and re-encoded:

$$\hat{\mathbf{x}}' = \mathcal{F}_\alpha^{(d)}(\hat{\mathbf{x}}), \quad \hat{\mathbf{z}} = E_\theta(\hat{\mathbf{x}}') \quad (5.5)$$

Latent Consistency Objective

Rather than enforcing pixel-level reconstruction which would force the model to attend to demographic appearance details we enforce *latent-space consistency*. The latent consistency loss comprises an ℓ_2 term and a cosine-similarity term:

$$\mathcal{L}_{\text{lat}} = \|\hat{\mathbf{z}} - \text{sg}(\mathbf{z})\|_2^2 + \beta (1 - \cos(\hat{\mathbf{z}}, \text{sg}(\mathbf{z}))), \quad \beta = 1.0 \quad (5.6)$$

where $\text{sg}(\cdot)$ denotes the *stop-gradient* operator (the target $\mathbf{z}$ is treated as a constant during the backward pass). By constraining the encoder to produce latents that survive a decode re-encode cycle, we encourage the representation to capture *semantic deepfake cues* (which persist through reconstruction) while discarding superficial appearance variations such as skin tone and facial structure (which are smoothed away by the decoder).

5.5 Adversarial Demographic Debiasing via Gradient Reversal

Demographic Classifier

A demographic classifier $D_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{G}|}$ maps the latent representation to a probability distribution over demographic groups. It is implemented as a two-layer MLP with Layer Normalisation and Max-Norm constraints ($c_{\text{max}} = 3.0$) to bound its complexity. The adversarial classification loss is:

$$\mathcal{L}_{\text{adv-clf}} = \mathbb{E}_{\mathbf{x}, g} [\text{CE}(D_\phi(\text{GRL}(\mathbf{z})), g)], \quad (5.7)$$

where $\text{GRL}(\cdot)$ is the Gradient Reversal Layer defined in Eq. 2.13.

Minimax Training Objective

The encoder and demographic classifier engage in a minimax game:

$$\min_{\theta} \max_{\phi} \mathcal{L}_{\text{AE}} = \lambda_{\text{lat}} \mathcal{L}_{\text{lat}}(\theta) + \lambda_{\text{fair}} \mathcal{L}_{\text{fair}}(\theta) - \lambda_{\text{adv}} \mathcal{L}_{\text{adv-clf}}(\theta, \phi). \quad (5.8)$$

The negative sign on $\mathcal{L}_{\text{adv-clf}}$ causes the encoder to *maximise* the adversary's classification error equivalently, to remove demographic information from $\mathbf{z}$. The GRL makes this

implicit: during the forward pass it acts as the identity; during back-propagation it negates the gradient by the reversal strength λ_{grl} (Eq. 2.13), avoiding the need for alternating optimisation.

5.6 CVaR Fairness Loss

Adversarial debiasing removes demographic information from $\mathbf{z}$ in an *average* sense, but may still leave vulnerable subgroups under-served. To protect these subgroups, we propose to augment the training objective with a *group-level CVaR* fairness loss (see Section 2.1.7):

$$\mathcal{L}_{\text{fair}} = \max_{y, a} \text{CVaR}_\alpha(\ell(\mathbf{x}, y) \mid y, a), \quad (5.9)$$

where $y \in \{0, 1\}$ is the real/fake label, a is the demographic group, and the inner CVaR is taken over the loss distribution of that demographic–label cell. Taking the max over all (y, a) combinations focuses optimisation on the *worst-performing subgroup*, ensuring Equalized Odds across the population. This connects directly to the group-level CVaR formulation of Eq. 4.9.

5.7 Model Pipeline

The proposed model pipeline ought to be trained in two stages:

Stage 1 : Fair Representation Learning

The adversarial autoencoder is to be trained end-to-end to learn demographic-agnostic semantically meaningful representations:

$$\theta^{(1)}, \phi^* = \arg \min_{\theta} \arg \max_{\phi} \mathcal{L}_{\text{AE}}(\theta, \phi). \quad (5.10)$$

Stage 2 : Binary Classification

Once the encoder has converged, E_θ is *frozen* and a separate real/fake classifier C_ψ ought to be trained on the fixed representations:

$$\psi^* = \arg \min_{\psi} \mathcal{L}_{\text{clf}}(\psi; \theta^{(1)}) = \mathbb{E}_{\mathbf{x}, y} [\text{CE}(C_\psi(\mathbf{z}), y)]. \quad (5.11)$$

Decoupling representation learning from classification prevents the classifier’s cross-entropy gradient from re-introducing demographic bias into $\mathbf{z}$.

5.8 Chapter Summary

This chapter presented FAIR-FRFT, our proposed fairness-aware deepfake detection framework. The key design decisions and their justifications are:

- **FrFT pre-processing** maps images into a hybrid spatial frequency domain where deepfake artefacts are more discriminative and less entangled with demographic appearance features.
- **Latent consistency training** replaces pixel reconstruction with a semantic cycle-consistency objective, preventing the autoencoder from memorising demographic appearance cues.
- **Gradient Reversal** implements adversarial debiasing within a single forward-backward pass, actively removing demographic information from the latent space without alternating optimisation.
- **CVaR fairness loss** complements adversarial debiasing by directly penalising worst-case subgroup performance, linking the training objective to the Equalized Odds fairness criterion.
- **Two-stage training** decouples fair representation learning from task-specific classification, preventing the classifier’s gradient from re-introducing bias.

Chapter 6

Conclusion, Future Work, and Industry Approaches

Declaration of Contributions: This chapter was compiled and authored by *Vishesh Gupta*, who interpreted the fairness metrics, analysed experimental results, and connected the technical findings with real-world AI systems deployed by industry leaders. *Nitin Agiwal* contributed the concluding synthesis of bias and fairness across the machine-learning pipeline. All team members cross-checked factual claims and references for accuracy.

This chapter consolidates the key findings of the report, outlines directions for future investigation, and situates the research within the broader landscape of how major AI organisations currently address algorithmic fairness. We begin with a synthesis of our conclusions, followed by concrete open problems, and close with a comparative survey of fairness approaches adopted by OpenAI, Google, and IBM.

6.1 Conclusion

Bias in AI is a systemic problem deeply embedded across the entire machine-learning pipeline, from data collection and model training through to evaluation. It arises due to historical inequalities reflected in data, underrepresentation of certain demographic groups, and human subjectivity in labelling and feature selection [29]. Real-world case studies such as the COMPAS recidivism algorithm and Amazon’s hiring system demonstrate that such biases can directly harm marginalised communities and perpetuate existing social inequalities at scale.

Traditional evaluation metrics such as accuracy are insufficient for assessing fairness:

models can achieve high overall performance while producing disproportionately harmful outcomes for specific demographic groups. This work examined the FD-VAE architecture [1] as a principled approach to fair representation learning. FD-VAE addresses the fundamental flaw in prior methods, most notably FFVAE, by introducing a three-way disentanglement of the latent space into *target* (z_t), *protected* (z_p), and *mutual* (z_m) attribute components, ensuring that the shared predictive information is exploited without leaking demographic signals into downstream classifiers.

The incorporation of Conditional Value-at-Risk (CVaR) loss [30] further strengthens robustness by focusing training on worst-performing subgroups rather than on average-case behaviour, encouraging fairness even in the absence of explicit group labels. The proposed pipeline extends these ideas to deepfake detection by combining Fractional Fourier Transform (FrFT) preprocessing [31], latent consistency enforcement, adversarial debiasing, and CVaR fairness loss. Together, these components aim to build a detection system that is not only accurate but equitable across demographic groups.

The ultimate goal is to develop AI systems that are **trustworthy**, **socially responsible**, and **fair by design** rather than as an afterthought.

6.2 Future Work

Several directions remain open for further investigation.

Empirical Validation. The proposed FrFT-based pipeline has been designed at an architectural level; empirical validation on standard benchmarks such as CelebA and UTK Face datasets is needed to quantitatively confirm its fairness–accuracy trade-offs against existing baselines including FFVAE and FD-VAE [1].

Multi-Attribute Fairness. The current approach assumes binary protected attributes such as gender. Extending the framework to handle multiple intersecting sensitive attributes simultaneously, race, age, and gender combined, presents a non-trivial challenge that warrants dedicated study. Intersectional fairness metrics that jointly penalise subgroup disparities would be a natural first step.

CVaR Hyperparameter Scheduling. The CVaR parameter α critically governs the fairness–accuracy trade-off [30]. A systematic sensitivity analysis and an adaptive scheduling strategy for α during training could improve generalisation across datasets with varying demographic distributions.

Training Stability of Adversarial Objectives. While adversarial debiasing is effective, it introduces the training instability common to minimax objectives. Exploring alternative

decorrelation strategies, such as kernel-based independence tests or information-theoretic regularisation, may yield more stable and scalable training dynamics.

Broader Application Domains. The applicability of this pipeline to high-stakes domains beyond deepfake detection, including medical imaging, credit scoring, and hiring systems, represents a promising and socially impactful direction. Each domain introduces unique distributional challenges (e.g. class imbalance in medical data, legally protected attributes in credit scoring) that would require careful adaptation of the proposed fairness constraints.

6.3 How AI Industry Leaders Address Fairness

Beyond academic research, major technology companies have developed their own fairness frameworks and tools. These approaches span training-time interventions, post-processing audits, and open-source toolkits. Three leading efforts are examined below.

6.3.1 OpenAI , Reinforcement Learning with Human Feedback (RLHF)

OpenAI employs Reinforcement Learning with Human Feedback (RLHF) [32] to align model behaviour with human values, including fairness. The key techniques include:

- **Human reviewers** who guide the model toward safe, unbiased responses through preference-based feedback.
- **Bias evaluation benchmarks** that test model outputs across demographic groups before deployment.
- **Content moderation layers** that filter outputs in real time to prevent discriminatory or toxic content.

The primary goal of RLHF in this context is to ensure that language models avoid generating discriminatory responses or perpetuating harmful stereotypes. Human oversight is central to this approach, making it effective but also resource-intensive and dependent on the diversity of human reviewers.

6.3.2 Google , Responsible AI Policies

Google has embedded fairness considerations into its model development lifecycle through its Responsible AI framework, applied prominently in the Gemini family of models [33]. The key techniques are:

- **Safety tuning during training**, where dataset composition and sampling strategies are adjusted to reduce demographic disparities before training begins.
- **Dataset filtering** to remove or rebalance data that may encode historical inequalities.
- **Fairness checks** that explicitly test for gender-, race-, and culture-based disparities in model outputs.

Google’s approach addresses bias at the source, during data preparation and model training, and is generally more effective than post-hoc correction. However, ensuring comprehensive coverage across all demographic dimensions remains an ongoing challenge.

6.3.3 IBM , AI Fairness 360 (AIF360)

IBM has released AI Fairness 360 (AIF360) [34], an open-source Python toolkit designed to help practitioners detect and mitigate bias across the full machine-learning pipeline. The toolkit provides:

- **Bias detection algorithms** that quantify disparate impact, statistical parity, and other fairness metrics.
- **Bias mitigation algorithms** categorised by intervention stage:
 - *Pre-processing*: Reweighting and disparate impact remover applied to training data to reduce bias at its source, ensuring that the model learns from a more balanced representation of the data.
 - *In-processing*: Adversarial debiasing and prejudice remover applied during model training, actively guiding the learning algorithm to reduce bias while it is being trained.
 - *Post-processing*: Calibrated equalized odds and reject option classification applied to model outputs, adjusting final predictions to ensure more equitable outcomes across different groups.

AIF360’s open-source nature makes it widely accessible and adaptable across industries. It supports a wide range of fairness metrics and can be integrated into existing machine-learning workflows with minimal overhead.

6.3.4 Comparative Summary

Table 6.1 summarises the three approaches along the dimensions of primary strategy, key techniques, and the types of harm each is designed to avoid.

Company	Approach	Techniques	What It Avoids
OpenAI (ChatGPT / GPT-4)	RLHF (Reinforcement Learning with Human Feedback) [32]	Human re-viewers guide safe responses; bias evaluation benchmarks; content moderation layers	Toxic outputs; discriminatory responses
Google (Gemini)	Responsible AI Policies [33]	Safety tuning during training; dataset filtering	Gender-, race-, and culture-based disparities (via fairness checks)
IBM (AIF360)	Open-source Fairness Toolkit [34]	Bias detection; bias mitigation algorithms at pre-, in-, and post-processing stages	Algorithmic bias across protected groups

Table 6.1: Comparison of industry approaches to AI fairness.

Bibliography

- [1] Seong Jae Park, Seong Min Kim, Sung Ju Hwang, and Hyeran Byun. Readme: Representation learning by fairness-aware disentangling method. *arXiv preprint arXiv:2007.03775*, 2020.
- [2] Yan Ju, Shu Hu, Shan Jia, George H. Chen, and Siwei Lyu. Improving fairness in deepfake detection, 2023.
- [3] McKinsey & Company. The state of ai: How organizations are rewiring to capture value, 2025.
- [4] Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments, 2017.
- [5] Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner. Machine bias: Risk assessments in criminal sentencing, 2016.
- [6] Jeffrey Dastin. Amazon scraps secret ai recruiting tool that showed bias against women, 2018.
- [7] Satoshi Watanabe. Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development*, 4(1):66–82, 1960.
- [8] R Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2:21–42, 2000.
- [9] X Ju et al. Fair deepfake detection via distributionally robust optimization. *arXiv preprint arXiv:2024.xxxxx*, 2024.
- [10] Robert C Williamson and Aditya Krishna Menon. Fairness risk measures. *International Conference on Machine Learning (ICML)*, pages 6786–6797, 2019.
- [11] Yoshua Bengio, Aaron Courville, and Pascal Vincent. Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8):1798–1828, 2013.

- [12] Irina Higgins, Loic Matthey, Arka Pal, et al. beta-vae: Learning basic visual concepts with a constrained variational framework. *International Conference on Learning Representations (ICLR)*, 2017.
- [13] Hyunjik Kim and Andriy Mnih. Disentangling by factorising. In *International Conference on Machine Learning (ICML)*, pages 2649–2658, 2018.
- [14] Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *International Conference on Learning Representations (ICLR)*, 2014.
- [15] Tian Qi Chen, Xuechen Li, Roger Grosse, and David Duvenaud. Isolating sources of disentanglement in variational autoencoders. *Advances in Neural Information Processing Systems (NeurIPS)*, 31, 2018.
- [16] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, et al. Generative adversarial nets. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 27, 2014.
- [17] Yaroslav Ganin and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.
- [18] Haldun M Ozaktas, Zeev Zalevsky, and M Alper Kutay. The fractional fourier transform. *Wiley*, 2001.
- [19] Xiaoqing Liu et al. Seeing is not believing: Deepfake detection using frequency domain analysis. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2021.
- [20] Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 29, 2016.
- [21] Sungho Park et al. Fairness-aware disentangling variational auto-encoder. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020.
- [22] Elliot Creager, David Madras, Toniann Pitassi, and Richard Zemel. Flexibly fair representation learning by disentanglement. *International Conference on Machine Learning (ICML)*, pages 1436–1445, 2019.
- [23] Elliot Creager, David Madras, Toniann Pitassi, and Richard Zemel. Flexibly fair representation learning by disentanglement. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2019.
- [24] Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *International Conference on Learning Representations (ICLR)*, 2014.

-
- [25] Satoshi Watanabe. Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development*, 4(1):66–82, 1960.
- [26] Hyunjik Kim and Andriy Mnih. Disentangling by factorising. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2018.
- [27] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015.
- [28] Zhifei Zhang, Yang Song, and Hairong Qi. Age progression/regression by conditional adversarial autoencoder. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [29] Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner. Machine bias: Risk assessments in criminal sentencing, 2016.
- [30] Ying Ju, Shu Hu, Shiqi Jia, Guang-Hong Chen, and Siwei Lyu. Improving fairness in deepfake detection. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 4655–4665, 2024.
- [31] Haldun M. Ozaktas and M. Alper Kutay. The fractional Fourier transform. In *2001 European Control Conference (ECC)*, pages 1477–1483. IEEE, September 2001.
- [32] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, and John Schulman. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, December 2022.
- [33] Gemini Team, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M. Dai, Anja Hauth, Katie Millican, David Silver, and Melvin Johnson. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- [34] Kalapriya Kannan, Pranay Kumar Lohia, Jacquelyn Martino, Sameep Mehta, Aleksandra Mojsilovic, Seema Nagar, Karthikeyan Natesan, John Richards, Diptikalyan Saha, Prasanna Sattigeri, and Moninder Singh. AI fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias. *IBM Journal of Research and Development*, 63(4/5):4:1–4:15, 2019.